

SingleRAN

SSL Feature Parameter Description

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Contents

1 Change History.....	1
1.1 SRAN22.1 Draft B (2026-01-30).....	1
1.2 SRAN22.1 Draft A (2025-12-31).....	1
2 About This Document.....	3
2.1 General Statements.....	3
2.2 Applicable RAT.....	3
2.3 Features in This Document.....	4
3 SSL.....	5
3.1 Principles.....	5
3.1.1 Overview.....	5
3.1.2 Procedure.....	5
3.1.2.1 SSL Protocol Stack.....	5
3.1.2.2 SSL Connection Establishment.....	6
3.1.2.3 SSL Renegotiation.....	10
3.1.2.4 SSL Key Update.....	13
3.1.2.5 SSL Cipher Suites.....	15
3.1.3 Application.....	20
3.1.3.1 OM Channel.....	21
3.1.3.1.1 OM Channel of the Base Station.....	21
3.1.3.1.2 OM Channel of the BSC6900/BSC6910 (GSM/UMTS).....	29
3.1.3.1.3 OM Channel of the BSC6960 (GSM/UMTS).....	30
3.1.3.2 FTP Transmission.....	34
3.1.3.3 HTTP Transmission.....	36
3.1.3.4 LDAP Transmission.....	37
3.1.3.5 eCPRI Interface Management Plane.....	40
3.2 Network Analysis.....	40
3.2.1 Benefits.....	40
3.2.2 Impacts.....	41
3.3 Requirements.....	41
3.3.1 Licenses.....	41
3.3.2 Functions.....	41
3.3.2.1 GBFD-113522 Encrypted Network Management.....	41

3.3.2.1.1 Prerequisite Functions.....	41
3.3.2.1.2 Mutually Exclusive Functions.....	42
3.3.2.1.3 Function Impacts.....	42
3.3.2.2 SSL on the NodeB Side.....	42
3.3.2.2.1 Prerequisite Functions.....	42
3.3.2.2.2 Mutually Exclusive Functions.....	42
3.3.2.2.3 Function Impacts.....	42
3.3.2.3 LBFD-004003 Security Socket Layer.....	42
3.3.2.3.1 Prerequisite Functions.....	42
3.3.2.3.2 Mutually Exclusive Functions.....	42
3.3.2.3.3 Function Impacts.....	43
3.3.2.4 TDLBFD-004003 Security Socket Layer.....	43
3.3.2.4.1 Prerequisite Functions.....	43
3.3.2.4.2 Mutually Exclusive Functions.....	43
3.3.2.4.3 Function Impacts.....	43
3.3.2.5 MLBFD-12000403 Security Socket Layer.....	43
3.3.2.5.1 Prerequisite Functions.....	43
3.3.2.5.2 Mutually Exclusive Functions.....	43
3.3.2.5.3 Function Impacts.....	43
3.3.2.6 SSL on the Base Station Controller Side.....	44
3.3.2.6.1 Prerequisite Functions.....	44
3.3.2.6.2 Mutually Exclusive Functions.....	44
3.3.2.6.3 Function Impacts.....	44
3.3.2.7 FBFD-010023 Security Mechanism (SSL).....	44
3.3.2.7.1 Prerequisite Functions.....	44
3.3.2.7.2 Mutually Exclusive Functions.....	44
3.3.2.7.3 Function Impacts.....	44
3.3.3 Hardware.....	44
3.3.4 Others.....	45
3.4 Operation and Maintenance.....	46
3.4.1 For the Base Station.....	46
3.4.1.1 When to Use.....	46
3.4.1.2 Data Configuration.....	46
3.4.1.2.1 Data Preparation.....	47
3.4.1.2.2 Using MML Commands.....	58
3.4.1.2.3 Using the MAE-Deployment.....	58
3.4.1.3 Activation Verification.....	59
3.4.1.4 Configuring the OM Channel on the MAE.....	62
3.4.1.5 Network Monitoring.....	62
3.4.2 For the BSC6900/BSC6910 (GSM/UMTS).....	62
3.4.2.1 When to Use.....	62
3.4.2.2 Data Configuration.....	62

3.4.2.2.1 Data Preparation.....	62
3.4.2.2.2 Using MML Commands.....	67
3.4.2.2.3 Using the MAE-Deployment.....	68
3.4.2.3 Activation Verification.....	68
3.4.2.4 Configuring the OM Channel of the BSC6900/BSC6910 on the MAE.....	69
3.4.2.5 Network Monitoring.....	69
3.4.3 For the BSC6960 (GSM/UMTS).....	69
3.4.3.1 When to Use.....	69
3.4.3.2 Data Configuration.....	69
3.4.3.2.1 Data Preparation.....	69
3.4.3.2.2 Using MML Commands.....	77
3.4.3.2.3 Using the MAE-Deployment.....	78
3.4.3.3 Activation Verification.....	78
3.4.3.4 Configuring the OM Channel of the BSC6960 on the MAE.....	81
3.4.3.5 Network Monitoring.....	82
4 Parameters.....	83
5 Counters.....	85
6 Glossary.....	86
7 Reference Documents.....	87

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical Changes
Changes in functions and their corresponding parameters
- Editorial Changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised descriptions of SSL reconnection between the MAE and base station or BSC6960. For details, see [3.1.3.1.1 OM Channel of the Base Station](#) and [3.1.3.1.3 OM Channel of the BSC6960 \(GSM/UMTS\)](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 03 (2025-12-14).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added the support for obtaining CRLs over HTTP. For details, see: 3.1.3 Application 3.1.3.3 HTTP Transmission 3.4.1.2 Data Configuration	Modified parameter: Added the enumerated value HTTP to the CRLTSK.CRLGETMET HOD (LTE eNodeB, 5G gNodeB) parameter.	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RA T	Feature ID	Feature Name	Chapter/Section
GS M	GBFD-113522	Encrypted Network Management	3 SSL
LTE FD D	LBFD-004003	Security Socket Layer	
LTE TD D	TDLBFD-004003	Security Socket Layer	
NB- IoT	MLBFD-12000403	Security Socket Layer	
NR	FBFD-010023	Security Mechanism (SSL)	

3 SSL

3.1 Principles

3.1.1 Overview

When operators' services are carried on open public IP networks, the transmitted data is prone to be tampered with, resulting in compromised data integrity. To address this issue, transmission security protocols need to be used to secure data transmission.

Secure Sockets Layer (SSL) is a protocol that provides end-to-end secure transmission channels to ensure the integrity of transmitted data and identify the confidentiality of end-to-end interconnection. SSL is used between the Transmission Control Protocol (TCP) layer and application layer. It provides security protection for Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Telecommunication Network Protocol (TELNET), and Lightweight Directory Access Protocol (LDAP).

The SSL protocol is the predecessor of Transport Layer Security (TLS). The related versions include SSL1.0, SSL2.0, SSL3.0, TLS1.0, TLS1.1, TLS1.2, and TLS1.3.

TLS1.3 is supported by Huawei base stations, BSC6960, MAE (when communicating with a Huawei base station or BSC6960), and LMT, not by BSC6900 or BSC6910. The following sections describe TLS1.2 and TLS1.3 in detail.

3.1.2 Procedure

3.1.2.1 SSL Protocol Stack

The SSL protocol stack consists of two protocol layers: record layer and handshake layer, as shown in [Figure 3-1](#).

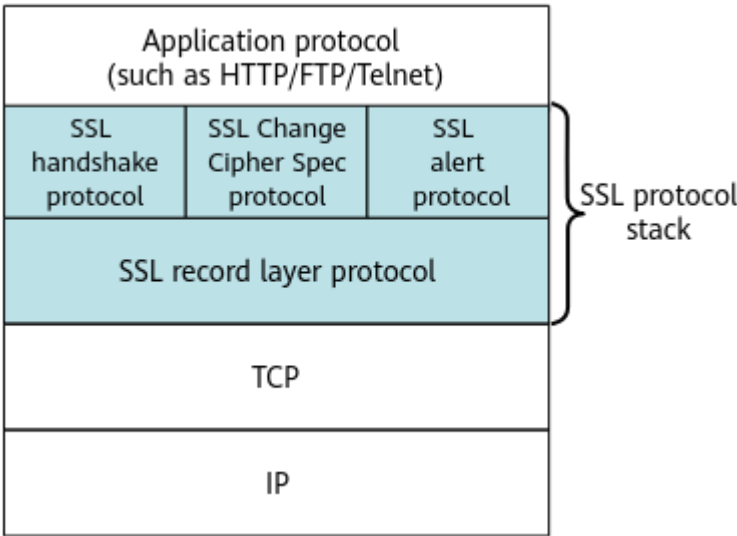
- Record layer

The record layer receives data from or transmits data to the application layer. In addition, the record layer performs security-related operations, such as

compression/decompression, encryption/decryption, and Medium Access Control (MAC) calculation.

- Handshake layer
The handshake layer consists of three protocols:
 - Handshake protocol
The handshake protocol establishes a security channel between the communicating parties before data transmission begins. During the handshake procedure, the communicating parties authenticate each other, negotiate encryption algorithms, generate keys, and generate initialization vectors (IVs).
 - Change Cipher Spec protocol
After the communicating parties agree on a set of new keys, each party sends a ChangeCipherSpec message to notify the other party that subsequent messages will be protected using the newly negotiated keys.
 - Alert protocol
An alert message indicates the severity of the alert. If there is a fatal alert message, the SSL connection is immediately terminated.

Figure 3-1 SSL protocol stack



3.1.2.2 SSL Connection Establishment

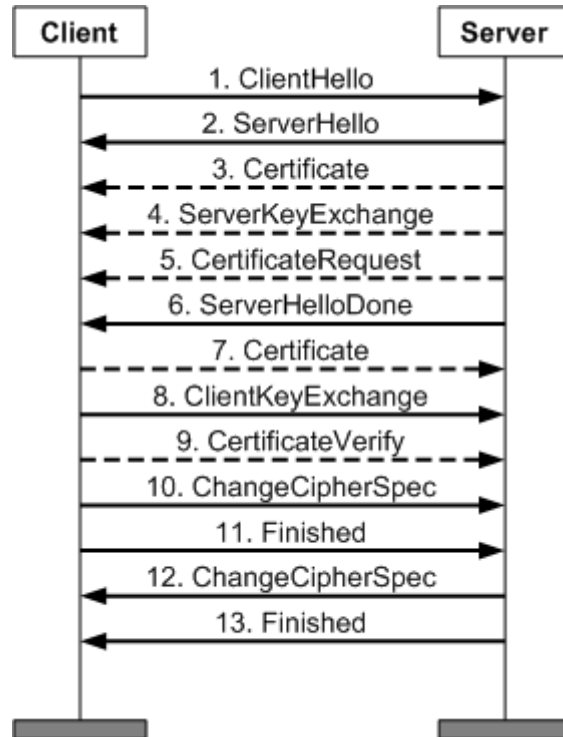
An SSL connection involves a handshake phase and a data transmission phase. Before data transmission, the client initiates an SSL handshake with the server. If the SSL handshake is successful, data is fragmented into manageable blocks that can be encrypted for transmission.

The purposes of the SSL handshake are as follows:

- The client and the server agree on a set of encryption algorithms, integrity check algorithms, and algorithm keys to secure data transmission.
- The communicating parties authenticate the identity of each other.

Figure 3-2 describes the general message exchange process between the client and the server during a TLS1.2 handshake.

Figure 3-2 General message exchange process between the client and the server during a TLS1.2 handshake



1. The client sends a ClientHello message to the server. This message contains the SSL version and a list of cipher suites supported by the client. The cipher suites are ranked by priority. For details about the cipher suites and their priorities, see [3.1.2.5 SSL Cipher Suites](#).
2. Upon receiving the ClientHello message, the server checks its cipher suite list for an identical cipher suite. If such a cipher suite is found, the server responds with a ServerHello message containing the SSL version and cipher suite selected by the server. If such a cipher suite is not found, the negotiation fails.
3. (Optional) The server sends a Certificate message containing its certificate to the client. If the negotiated algorithms require server authentication, the client uses this message for authentication.
4. (Optional) The server will send a ServerKeyExchange message immediately after the Certificate message to supplement the key information only if the key contained in the Certificate message is not long enough.
5. (Optional) If authenticating the client is required, the server sends a CertificateRequest message to the client.
6. The server sends a ServerHelloDone message to the client, notifying the client that the handshake is complete.
7. (Optional) If the client receives a CertificateRequest message from the server, the client sends a Certificate message containing its certificate to the server.
8. The client sends a ClientKeyExchange message to the server. This message contains the data used to generate the keys for encryption algorithms and

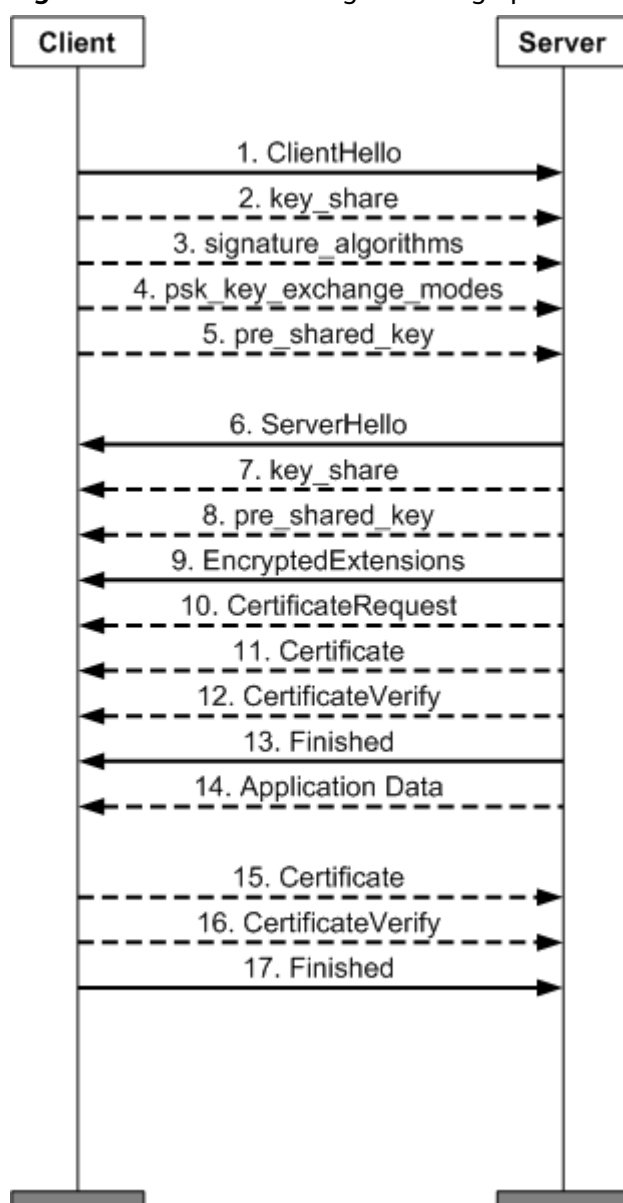
integrity check algorithms. If the client receives a ServerKeyExchange message from the server, the data used to generate the keys is encrypted using the key information carried in the ServerKeyExchange message.

9. (Optional) If the client receives a CertificateRequest message from the server, the client sends a CertificateVerify message signed by the private key associated with its certificate to the server.
10. The client sends a ChangeCipherSpec message to the server, notifying the server that the client will use the negotiated algorithms and keys for subsequent communications.
11. The client sends a Finished message to the server. This is the first message protected using the negotiated algorithms. This message contains the MAC of all messages transmitted during the handshake. The MAC is used to check whether handshake messages have been manipulated during transmission.
12. The server sends a ChangeCipherSpec message to the client. The function of this message is the same as that in 10.
13. The server sends a Finished message to the client. The function of this message is the same as that in 11.

After the handshake phase is complete, the client and the server begin to transmit data under SSL protection.

The handshake process of TLS1.3 is different from that of TLS1.2. TLS1.2 requires two round-trip times (RTTs) to complete the handshake. The first RTT contains ClientHello and ServerHello, and the second RTT contains ServerKeyExchange and ClientKeyExchange. TLS1.3 simplifies the key negotiation algorithm and requires only one RTT, greatly shortening the handshake delay. [Figure 3-3](#) shows the general message exchange process during a TLS1.3 handshake.

Figure 3-3 General message exchange process during a TLS1.3 handshake



1. The client sends a ClientHello message to the server. This message contains the SSL version and a list of cipher suites supported by the client. The cipher suites are ranked by priority. For details about the cipher suites and their priorities, see [3.1.2.5 SSL Cipher Suites](#).
2. (Optional) The client sends the key material and public key to the server through a key_share message.
3. (Optional) The client sends signature algorithms to the server through a signature_algorithms message.
4. (Optional) If the client needs to use the pre-shared key for negotiation, it sends the pre-shared key usage mode to the server through a psk_key_exchange_modes message.
5. (Optional) If the client needs to use the pre-shared key for negotiation, it sends a pre-shared key ID list to the server through a pre_shared_key message.

6. Upon receiving the ClientHello message, the server checks its cipher suite list as well as that of the client for an identical cipher suite. If such a cipher suite is found, the server responds with a ServerHello message. If such a cipher suite is not found, the negotiation fails.
7. (Optional) The server sends the key material and public key to the client through a key_share message.
8. (Optional) The server sends a pre-shared key ID list to the client through a pre_shared_key message.
9. The server sends an EncryptedExtensions message to the client. This message is encrypted using the negotiated key and contains TLS1.3 extension functions.
10. (Optional) If authenticating the client is required, the server sends a CertificateRequest message to the client.
11. (Optional) If authenticating the server is required, the server sends a Certificate message containing its certificate to the client.
12. (Optional) If authenticating the server is required, the server sends a CertificateVerify message signed by the private key associated with its certificate to the client.
13. The server sends a Finished message to the client. This is the first message protected using the negotiated algorithms. This message contains the MAC of all messages transmitted during the handshake. The MAC is used to check whether handshake messages have been manipulated during transmission.
14. (Optional) The server sends an Application Data message (application-layer data) to the client.
15. (Optional) If the client receives a CertificateRequest message from the server, the client sends a Certificate message containing its certificate to the server.
16. (Optional) If the client receives a CertificateRequest message from the server, the client sends a CertificateVerify message signed by the private key associated with its certificate to the server.
17. The client sends a Finished message to the server. The function of this message is the same as that in [13](#).

For details about SSL, see:

- RFC 5246 for TLS1.2
- RFC 8446 for TLS1.3
- RFC 5246 and RFC 5746 for TLS renegotiation

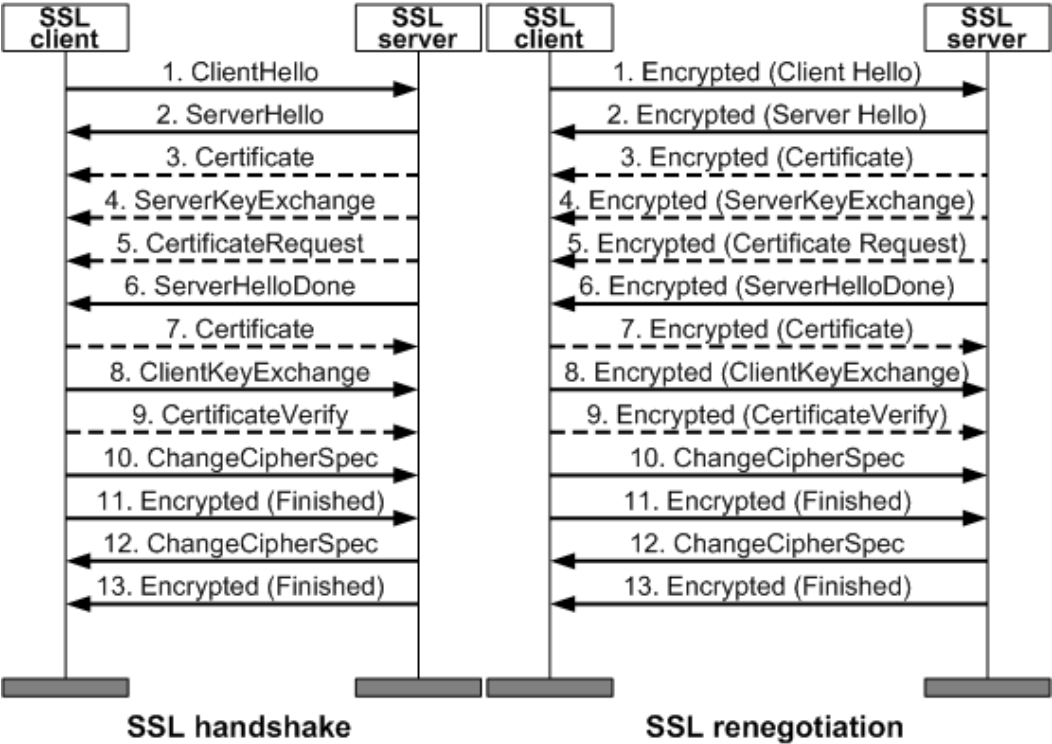
3.1.2.3 SSL Renegotiation

SSL Renegotiation Attack

SSL key renegotiation (SSL renegotiation for short) is used to negotiate a new cipher key between the SSL communicating parties. The new cipher key will replace the old one to avoid key disclosure after a long period of usage. This function applies only to TLS1.2.

The SSL renegotiation procedure is similar to the SSL handshake procedure, as shown in [Figure 3-4](#).

Figure 3-4 Comparison between the SSL renegotiation procedure and the SSL handshake procedure



In [Figure 3-4](#), the left part shows the SSL handshake procedure and the right part shows the SSL renegotiation procedure. The same signaling messages are transmitted during these two procedures. However, the messages shown as "Encrypted (message name)" are encrypted, while all other messages are transmitted in plaintext. According to SSL protocols, the ChangeCipherSpec message is the only message that is transmitted in plaintext. Other renegotiation messages are transmitted in encrypted channels.

According to RFC 5246, an SSL renegotiation procedure is proactively initiated by one end on an SSL connection. The other end passively accepts the SSL renegotiation request. This procedure generates a new encryption key. During this procedure, the SSL client and server use different amount of computational resources. Generally, the computational resources used by the SSL server are about ten times those used by the SSL client during an SSL renegotiation procedure.

If a malicious SSL client frequently sends SSL renegotiation requests to the SSL server, the SSL server will consume a large amount or possibly all of the resources. As a result, other SSL clients cannot initiate SSL connections to the SSL server. This attack is called a denial-of-service (DoS) attack. In the worst case, a DoS attack will cause the SSL server to break down due to resource exhaustion.

In an SSL connection, a Huawei base station controller/base station usually acts as the SSL server and the peer end acts as the SSL client.

SSL Renegotiation Management

The principle of SSL renegotiation management is as follows: The **SSL.RENEGO (LTE eNodeB, 5G gNodeB)** and **SSL.RENEGOINTERVAL (LTE eNodeB, 5G**

gNodeB) parameters are configured on the SSL server (for example, base station controller/base station) to limit the number of SSL renegotiation requests sent by SSL clients. This configuration reduces the resource consumption on the SSL server and prevents DoS attacks. The **SSL.RENEGO (LTE eNodeB, 5G gNodeB)** parameter specifies whether to allow SSL renegotiation, and the **SSL.RENEGOINTERVAL (LTE eNodeB, 5G gNodeB)** parameter specifies the interval at which SSL renegotiation is initiated. For example, when the base station controller/base station acts as the SSL server, renegotiation can be configured and managed through the following parameter settings.

SSL Server Side	SSL.RENEGO (LTE eNodeB, 5G gNodeB)	SSL.RENEGOINTERVAL (LTE eNodeB, 5G gNodeB)	Rule
Both SSL renegotiation and SSL renegotiation management are disabled.	DISABLE	Not configured	The SSL server rejects all SSL renegotiation requests sent by the SSL client and returns warning-level Alert messages with the cause value "no_renegotiation(100)" to the SSL client. For version compatibility concerns, the SSL server will not proactively disconnect the current SSL connection after the SSL renegotiation rejection. According to specifications, upon receiving a warning-level renegotiation rejection message, the SSL client continues using the current SSL connection or disconnects it based on the configuration of the SSL client.
SSL renegotiation is enabled, and SSL renegotiation management is disabled.	ENABLE	0	Whenever the SSL client sends an SSL renegotiation request to the SSL server, the SSL server will accept and process the request and then renegotiate a cipher key if the request complies with the protocol.

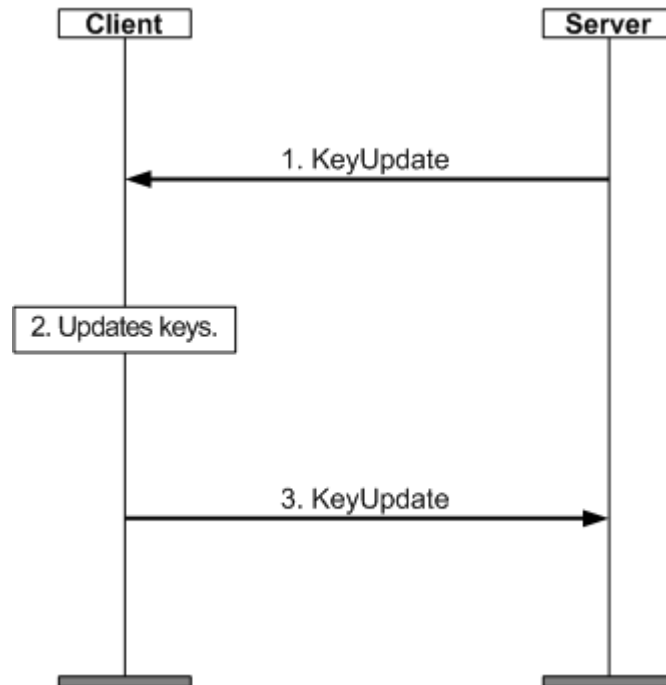
SSL Server Side	SSL.RENEGOTIATION (LTE eNodeB, 5G gNodeB)	SSL.RENEGOTIATION_INTERVAL (LTE eNodeB, 5G gNodeB)	Rule
Both SSL renegotiation and SSL renegotiation management are enabled.	ENABLE	X (ranging from 1 to 60)	The SSL server accepts and processes SSL renegotiation requests sent by the SSL client, and the interval between two valid SSL renegotiation requests must be longer than X minutes. When the SSL client sends the first SSL renegotiation request message to the SSL server, the SSL server processes the request. The SSL server then generates a new cipher key if it determines that the request message is valid. Subsequently, the following may occur when the SSL client sends another SSL renegotiation request: • If the interval between the two SSL renegotiation requests is shorter than or equal to X minutes, the SSL server rejects the latter SSL renegotiation request and returns a warning-level Alert message with the cause value "no_renegotiation(100)" to the SSL client. For version compatibility concerns, the SSL server will not proactively disconnect the current SSL connection after the SSL renegotiation rejection. • If the interval between the two SSL renegotiation requests is longer than X minutes, the SSL server accepts and processes the latter SSL renegotiation request and renegotiates a cipher key.

3.1.2.4 SSL Key Update

In TLS1.3, a key update function is introduced to replace SSL renegotiation. Different from SSL renegotiation, the SSL key update function uses a key derivation function to update the key that is being used by the current connection. With this method, asymmetric encryption is not involved on the server and client, and only a few computational resources are consumed. Therefore, this method will not cause system breakdown due to resource exhaustion.

The SSL key update procedure is performed after the SSL handshake is complete. According to RFC 8446, both the client and server can initiate an SSL key update procedure. **Figure 3-5** shows an SSL key update procedure initiated by the server.

Figure 3-5 SSL key update



1. The server sends a KeyUpdate message to notify the client of the key update mode. The key update modes include sending key update, sending/receiving key update, and not updating key. In addition, the server uses the key derivation function to update the key on the server.
2. Upon reception of the KeyUpdate message, the client uses the key derivation function to update the key on the client based on the key update mode contained in the message.
3. If the server's key update mode is sending key update, the client may send a KeyUpdate message according to the client's security policy, to request the server to update the server's receiving key. If the server's key update mode is sending/receiving key update, the client sends a KeyUpdate message in which the key update mode is not updating key.

The SSL key update management function is controlled by the **TLSPOLICY.KEYUPDATESW** and **TLSPOLICY.KEYUPDATELT** parameters, which are used to enable or disable the key update function and set the key update period, respectively. The following table describes the configurations in two scenarios.

Configuration Scenario	Key Update Switch	Key Update Lifetime	Rule
The SSL key update function and the SSL key update management function are disabled.	OFF	Not configured	The base station will not proactively initiate a key update procedure but can process the key update messages sent by the peer end.
The SSL key update function and the SSL key update management function are enabled.	ON	X (5–10080)	The base station proactively initiates a key update procedure every <i>X</i> minutes.

3.1.2.5 SSL Cipher Suites

During an SSL handshake procedure, the server and the client agree on a set of algorithms such as encryption algorithms, signature algorithms, key exchange algorithms, and MAC algorithms. For details about the SSL handshake procedure, see [3.1.2.2 SSL Connection Establishment](#). These algorithms are collectively called SSL encryption algorithm suites (cipher suites for short).

TLS1.2 Cipher Suites

NOTICE

For TLS1.2, Huawei base stations support only the extension of the uncompressed point format for elliptic curves. Therefore, if an ECDSA certificate uses the point format for its public key elliptic curve, the TLS connection will be interrupted.

In compliance with TLS1.2, Huawei products are preconfigured with SSL cipher suites of different security levels before delivery, as listed in [Table 3-1](#). A smaller cipher suite number (No.) indicates a higher priority. The cipher suite used by the SSL connection is supported on both sides and has the highest priority on the server side.

Table 3-1 Factory settings of the security levels and priorities of cipher suites

No.	SSL Cipher Suite	Security Level	Applicable RAT
01	ECDHE_ECDSA_AES_256_GCM_SHA384	HIGH	GSM, UMTS, LTE, and NR
02	ECDHE_RSA_AES_256_GCM_SHA384	HIGH	GSM, UMTS, LTE, and NR
03	ECDHE_ECDSA_AES_128_GCM_SHA256	HIGH	GSM, UMTS, LTE, and NR
04	ECDHE_RSA_AES_128_GCM_SHA256	HIGH	GSM, UMTS, LTE, and NR
05	DHE_RSA_AES_256_GCM_SHA384	HIGH	GSM, UMTS, LTE, and NR
06	DHE_RSA_AES_128_GCM_SHA256	HIGH	GSM, UMTS, LTE, and NR

 NOTE

Base station controllers, base stations, and MAEs do not necessarily support the SSL cipher suites listed in [Table 3-1](#) by default. Different types of NEs may support different cipher suites.

The cipher suites in [Table 3-1](#) are all secure and are sorted based on the algorithm security strength. Users are advised to preferentially enable the AES_256_GCM_SHA384 cipher suite, which has a high security level.

The authentication algorithm type in the negotiated cipher suite must be consistent with the signature algorithm type in the device certificate of the base station/base station controller. If they are inconsistent, the SSL cipher suite cannot be used.

For example, if the authentication algorithm type in the cipher suite ECDHE_ECDSA_AES_256_GCM_SHA384 is ECDSA and the signature algorithm type in the device certificate of the base station is RSA, this cipher suite cannot be used.

The following SSL/TLS cipher suites are insecure or have security vulnerabilities and have been blacklisted by HTTP2.0: DHE_RSA_AES_256_CBC_SHA256, DHE_RSA_AES_256_CBC_SHA, DHE_RSA_AES_128_CBC_SHA256, and DHE_RSA_AES_128_CBC_SHA. For details, see RFC 7540.

To meet compatibility requirements, the following anonymous authentication algorithms are included in the cipher suite with a high security level by default:

- DH_anon_AES_256_CBC_SHA256
- DH_anon_AES_256_CBC_SHA

The anonymous authentication cipher suite with a high security level takes effect only when the following conditions are met: (1) The **SSL.AUTHMODE (LTE eNodeB, 5G gNodeB)** parameter of the base station/BSC6960 is set to **NONE**. (2) The base station/BSC6960 does not have a device certificate or private key file, or certificate loading fails.

The preceding anonymous authentication algorithms are automatically disabled if either of the following conditions is met:

- The **SSL.AUTHMODE (LTE eNodeB, 5G gNodeB)** parameter of the base station/BSC6960 is set to **NONE**, and a device certificate or private key file has been successfully loaded to the base station/BSC6960.
- The **SSL.AUTHMODE (LTE eNodeB, 5G gNodeB)** parameter of the base station/BSC6960 is set to **PEER**.

 NOTE

The default value of the **SSL.AUTHMODE (LTE eNodeB, 5G gNodeB)** parameter for Huawei BSC6960s, eGBTSs, NodeBs, eNodeBs, and gNodeBs is **PEER**. When an anonymous authentication algorithm is required to be enabled, the value of this parameter must be modified to **NONE** on the LMT to restore the OM channel.

If the base station controller/MAE acts as the SSL server, anonymous authentication algorithms cannot be automatically added to cipher suites of high security levels.

The security levels of the cipher suites supported by the base station controller/base station can be adjusted by changing the setting of the **SSLCS** MO, realizing the evolution of security for the cipher suites.

 NOTE

If the base station acts as the SSL client and the client version is TLS1.2 or earlier, its cipher suites are fixed and cannot be configured.

The following uses the base station controller/base station acting as the SSL server, the MAE acting as the SSL client, and the cipher suite ECDHE_RSA_AES_256_GCM_SHA384 as an example. [Table 3-2](#) describes the configuration of the **SSLCS** MO for the SSL server.

Table 3-2 Configuration of the **SSLCS** MO

SSLCS.CSNAME Value	SSLCS.CSLEVEL Value	Description
ECDHE_RSA_AES_256_GCM_SHA384	DISABLE	Indicates that the cipher suite ECDHE_RSA_AES_256_GCM_SHA384 is disabled at the SSL server and this cipher suite is not used for SSL negotiations. SSL connections that are using this cipher suite will be disconnected automatically.

SSLCS.CSNAME Value	SSLCS.CSLEVEL Value	Description
	LOW	Indicates a low security level of the SSL server cipher suite ECDHE_RSA_AES_256_GCM_SHA384. NOTE When the security level of a cipher suite is changed, SSL connections that are using this cipher suite will not immediately become invalid. The new security level of this cipher suite will take effect only after an SSL renegotiation is initiated or a new SSL connection is established.
	MEDIUM	Indicates a medium security level of the SSL server cipher suite ECDHE_RSA_AES_256_GCM_SHA384. NOTE When the security level of a cipher suite is changed, SSL connections that are using this cipher suite will not immediately become invalid. The new security level of this cipher suite will take effect only after an SSL renegotiation is initiated or a new SSL connection is established.
	HIGH	Indicates a high security level of the SSL server cipher suite ECDHE_RSA_AES_256_GCM_SHA384. NOTE When the security level of a cipher suite is changed, SSL connections that are using this cipher suite will not immediately become invalid. The new security level of this cipher suite will take effect only after an SSL renegotiation is initiated or a new SSL connection is established.

The following example uses the base station controller or base station acting as the SSL server. The scope of cipher suites allowed by the SSL server during SSL negotiations is determined based on the configuration of the lowest-security-level parameter **SSLCONF.LOWESTCSLEVEL** (for the BSC6900, BSC6910, or GBTS) or **SSL.LOWESTCSLEVEL (5G gNodeB, LTE eNodeB)** (for the BSC6960, gNodeB, eNodeB, NodeB, or eGBTS) on the SSL server, as described in [Table 3-3](#), to meet security and compatibility requirements of specific scenarios.

Table 3-3 Cipher suites allowed by the SSL server configured with the lowest-security-level parameter

LOWESTCSLEVEL Value	LOW	MEDIUM	HIGH
Cipher Suite at High Security Level	Allowed	Allowed	Allowed

Cipher Suite at Medium Security Level	Allowed	Allowed	Not allowed
Cipher Suite at Low Security Level	Allowed	Not allowed	Not allowed
Disabled Cipher Suite	Not allowed	Not allowed	Not allowed

TLS1.3 Cipher Suites

According to RFC 8446, TLS1.3 cipher suites are different from TLS1.2 cipher suites. TLS1.3 cipher suites are a collection of Diffie-Hellman groups, signature algorithms, and cipher suites. [Table 3-4](#), [Table 3-5](#), and [Table 3-6](#) list TLS1.3 algorithms supported by base stations.

Table 3-4 Diffie-Hellman groups

No.	Diffie-Hellman Group
01	X448
02	X25519
03	SECP384R1
04	SECP256R1

NOTICE

The setting of the **TLSPOLICY.DHGRP (5G gNodeB, LTE eNodeB)** parameter affects the verification of the TLS1.2 certificate signature algorithm. If the algorithm represented by an option of this parameter is disabled, connections using TLS1.2 may fail to be set up. Therefore, do not disable any algorithm represented by the options of this parameter if no security issue arises.

Table 3-5 Signature algorithms

No.	Signature Algorithm
01	RSA_PSS_RSAE_SHA256
02	RSA_PSS_RSAE_SHA384
03	ECDSA_SECP256R1_SHA256
04	ECDSA_SECP384R1_SHA384
05	ED25519

 NOTE

The **TLSPOLICY.SIGALG (5G gNodeB, LTE eNodeB)** parameter takes effect for both TLS1.2 and TLS1.3.

Table 3-6 Cipher suites

No.	Cipher Suite
01	TLS_AES_256_GCM_SHA384
02	TLS_CHACHA20_POLY1305_SHA256
03	TLS_AES_128_GCM_SHA256

The **TLSPOLICY** MO determines whether TLS1.3 algorithms for base stations are enabled. [Table 3-7](#) uses the TLS1.3 key negotiation algorithm X25519 as an example to describe the parameter settings in the **TLSPOLICY** MO.

 NOTE

The following TLS1.3 algorithms cannot be disabled: the Diffie-Hellman group SECP256R1, the signature algorithm RSA_PSS_RSAE_SHA256, and the cipher suite TLS_AES_128_GCM_SHA256

Table 3-7 Example of parameter settings in the **TLSPOLICY** MO

TLSPOLICY.DHGRP (5G gNodeB, LTE eNodeB) Value	Description
X25519 X448 SECP256R1 SECP384R1	The key negotiation algorithms X25519, X448, SECP256R1, and SECP384R1 are enabled.
X448 SECP256R1 SECP384R1	The X25519 algorithm is disabled. SSL connections that are using this algorithm will be disconnected automatically. The key negotiation algorithms X448, SECP256R1, and SECP384R1 are still enabled.

3.1.3 Application

The applications of SSL are listed in the following table.

Table 3-8 SSL application scenarios

NE	Base Station	Base Station Controller
MAE	OM channel/FTP connection	OM channel/FTP connection
LMT	HTTP connection	HTTP connection
Certification Authority (CA)	HTTP connection	HTTP connection
Certificate revocation list (CRL)	Lightweight Directory Access Protocol (LDAP)/FTP/HTTP connection	LDAP/FTP connection
AAU	eCPRI interface management plane	N/A

3.1.3.1 OM Channel

SSL protects data transmission on the OM channel between the base station/base station controller and the MAE (including optional components of the MAE, such as a trace server).

NOTE

By default, Telnet services on port 23 are disabled on the base station and base station controller. Port 6000 is used for TCP connections on both the base station and base station controller. If a client connects to the base station or base station controller through port 6000, only plaintext MML commands are allowed.

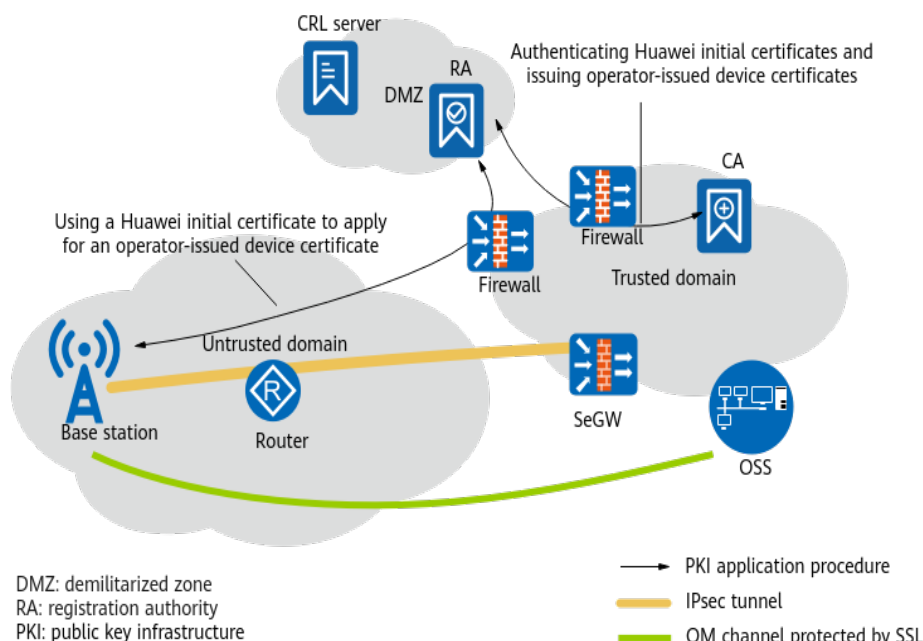
On the MAE-Access, to disable TCP connections in plaintext on port 6000, run the **SET CONNTYPE** command with the parameters set as follows:

- Base station controller other than the BSC6960: Set **CONNTYPE.CONNTYPE** to **ONLY_SSL**.
- gNodeB/eNodeB/NodeB/eGBTS/BSC6960: Set **SSL.CONNTYPE (LTE eNodeB, 5G gNodeB)** to **SSL**.

3.1.3.1.1 OM Channel of the Base Station

Figure 3-6 illustrates a typical network topology in which SSL is applied to the OM channel between the base station and the MAE. In this network topology, Internet Protocol Security (IPsec) is not applied to the OM channel.

Figure 3-6 Network topology for SSL applied to the OM channel between the base station and the MAE



Before configuring SSL, set the connection type between the MAE and base station to SSL and set the authentication mode to "verify peer certificate" on the MAE. In addition, preconfigure the operator-issued device certificate and the operator's root certificate on the MAE.

NOTE

When the **SSL.CONNTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **SSL** on the base station side, the base station needs to obtain the operator-issued device certificate and the operator's root certificate from the operator's public key infrastructure (PKI) system before establishing an SSL connection. For detailed operations, see *PKI*.

The process of establishing an SSL connection occurs as follows:

1. The base station and the MAE establish a TCP connection.
2. The MAE functions as an SSL client and initiates an SSL handshake with the base station.
3. During the SSL handshake, the base station and the MAE negotiate SSL versions, cipher suites, and an authentication mode. The MAE authenticates the base station using the negotiated authentication mode. Whether the base station authenticates the MAE depends on the configuration of the base station. After the authentication is successful, the base station and the MAE establish an OM channel protected by SSL.

NOTE

During base station deployment by PnP, the MAE can determine whether to authenticate the base station while the base station authenticates the peer end by default. Only the peer authentication mode is supported by the base station during the deployment and the setting cannot be changed.

When an OM channel is protected by IPsec, the process of establishing an SSL connection on the OM channel is the same as the previous process.

The SSL authentication mode of the OM channel between the base station and MAE is determined based on the configurations on both sides, as described in [Table 3-9](#).

Table 3-9 SSL authentication mode of the OM channel between the base station and the MAE

SSL Authentication Mode	Applicable RAT	Authentication Mode on the MAE Side	Authentication Mode (SSL.AUTHMODE Setting) on the Base Station Side	Deployment Requirement	Application Scenario
Non-authentication between the base station and MAE	GSM, UMTS, LTE, and NR	Anonymous	NONE	None	Routine maintenance Not supported during PnP deployment
Unidirectional authentication of the base station by the MAE	GSM, UMTS, LTE, and NR	Authentication of the NE by the OSS	NONE	The base station is preconfigured with an operator-issued device certificate and the operator's root certificate. The MAE is preconfigured with the operator's root certificate.	Routine maintenance Not supported during PnP deployment
Bidirectional authentication between the base station and MAE	GSM, UMTS, LTE, and NR	Bidirectional	PEER	Both the base station and MAE are preconfigured with an operator-issued device certificate and the operator's root certificate.	Routine maintenance PnP deployment

SSL Authentication Mode	Applicable RAT	Authentication Mode on the MAE Side	Authentication Mode (SSL.AUTHMODE Setting) on the Base Station Side	Deployment Requirement	Application Scenario
Unidirectional authentication of the MAE by the base station	GSM, UMTS, LTE, and NR	Authentication of the OSS by the NE	PEER	The base station is preconfigured with the operator's root certificate. The MAE is preconfigured with an operator-issued device certificate and the operator's root certificate.	Routine maintenance PnP deployment

When the PKI system is deployed on an operator's network, it is recommended that the base station and the MAE use the operator's certificates to authenticate each other. In this case, set the **APPCERT.APPTYPE (5G gNodeB, LTE eNodeB)** parameter to **SSL** and set the **APPCERT.APPCERT (5G gNodeB, LTE eNodeB)** parameter to the device certificate used for SSL authentication.

Table 3-10 describes the configurations when the operator uses the SSL PSK authentication mode.

Table 3-10 SSL PSK authentication mode of the OM channel between the base station and the MAE

Authentication Mode on the MAE Side	Applicable RAT	Authentication Mode (SSL.AUTHMETH (LTE eNodeB, 5G gNodeB) Setting) on the Base Station Side	Deployment Requirement	Application Scenario
SSL+PSK	GSM, UMTS, LTE, and NR	CERT_PRIORITY	The PSK is required on the base station and MAE.	Routine maintenance PnP deployment

When a base station works as an SSL client or server, certificates in the SSL trust group specified by the **SSL.TRUSTGROUPID (LTE eNodeB, 5G gNodeB)** parameter can be used to verify the device certificates of the peer end. If this parameter is not specified, all trust certificates of the base station are used to verify the device certificates of the peer end by default.

 NOTE

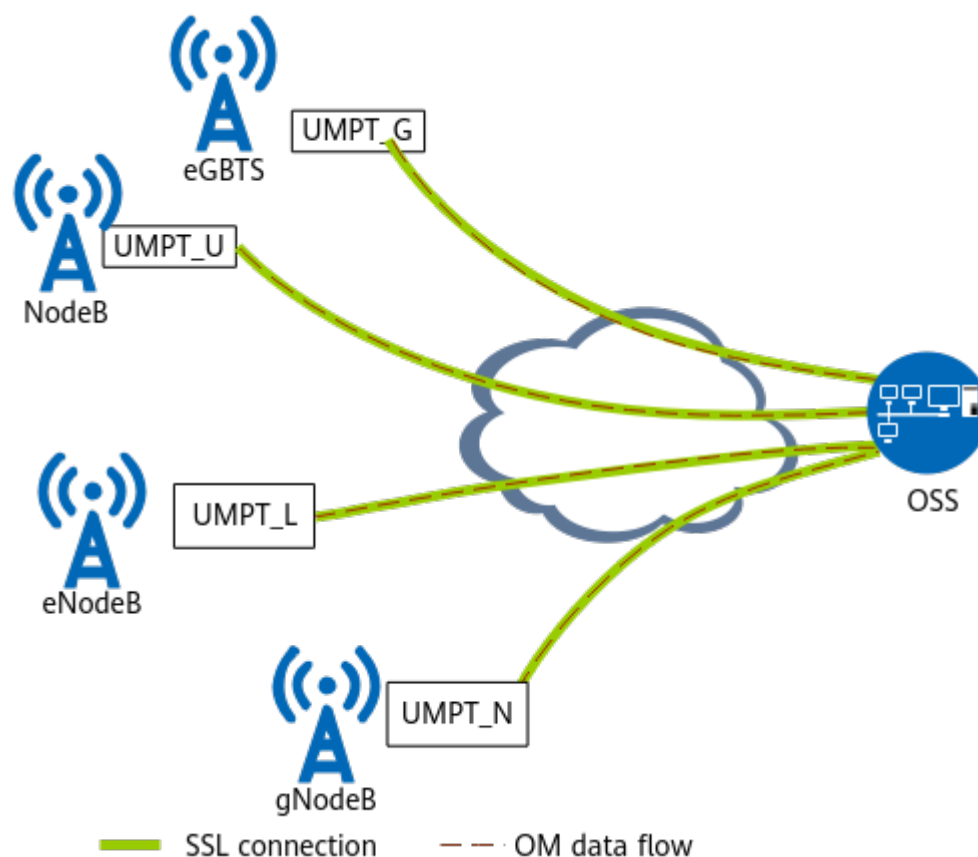
If the **SSL.AUTHMETH (LTE eNodeB, 5G gNodeB)** parameter is set to **CERT_PRIORITY**, the cipher suite TLS_AES_256_GCM_SHA384 cannot be disabled for the **TLSPOLICY.CIPHERSUITE (5G gNodeB, LTE eNodeB)** parameter. This constraint does not apply when the **SSL.AUTHMETH (LTE eNodeB, 5G gNodeB)** parameter is set to **CERT_ONLY**.

For details about trust certificates and trust groups, see *PKI*.

OM Channel of a Single-Mode Base Station

Figure 3-7 shows a network topology in which SSL is applied to the OM channel of an eGBTS, NodeB, eNodeB, or gNodeB. SSL is not applied to a GBTS. This is because SSL is based on the TCP protocol, whereas the OM data of a GBTS is encapsulated into UDP packets.

Figure 3-7 Network topology for SSL applied to the OM channel of a single-mode base station

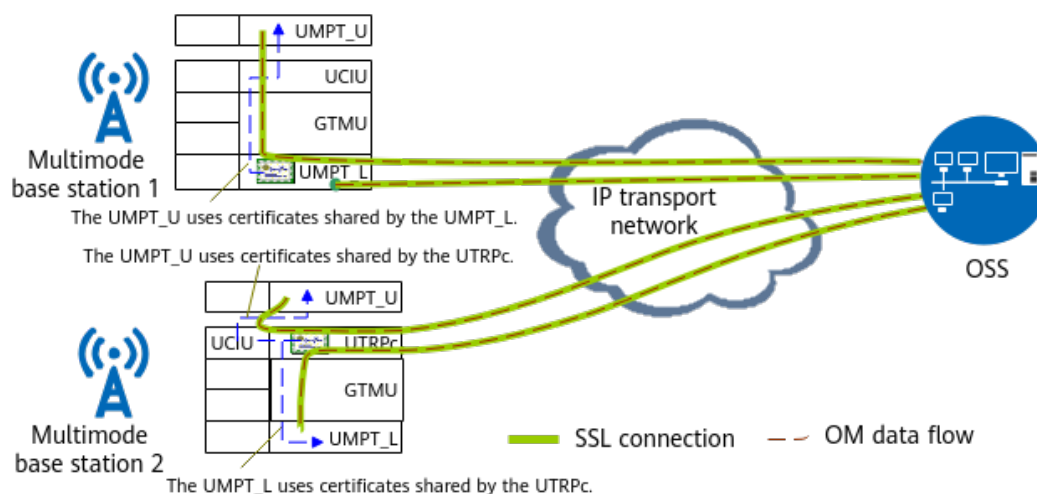


OM Channel of a Separate-MPT Multimode Base Station

When SSL is applied to the OM channels of a separate-MPT multimode base station, an SSL connection must be established between each RAT and the MAE. If a base station of a certain RAT requires SSL certificate authentication and no certificates are configured on the main control board of the RAT, this main control board must obtain shared certificates from another board through the backplane.

Figure 3-8 illustrates a certificate sharing example in which different RATs of a separate-MPT GUL base station share the same IPsec tunnel. The operator's certificates of multimode base station 1 are deployed on the UMPT_L. If an SSL connection needs to be established between the NodeB and MAE using the operator's certificates for authentication, the UMPT_U needs to obtain certificates from the UMPT_L through the backplane. The operator's certificates of multimode base station 2 are deployed on the UTRPc. If two SSL connections need to be established between the NodeB and MAE and between the eNodeB and MAE using the operator's certificates for authentication, the UMPT_U and UMPT_L need to obtain certificates from the UTRPc through the backplane.

Figure 3-8 Network topology for SSL applied to the OM channels of a separate-MPT GUL base station



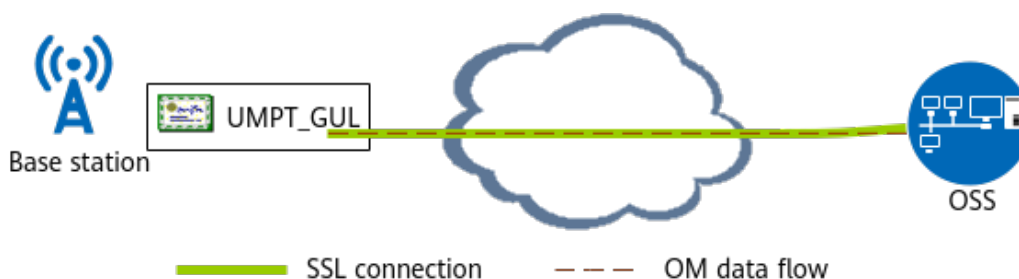
NOTE

A UMDU/MDUC cannot be used in a separate-MPT base station.

OM Channel of a Co-MPT Multimode Base Station

When SSL is applied to the OM channel of a co-MPT multimode base station, there is only one OM channel of the base station, as illustrated in [Figure 3-9](#). In this scenario, the SSL function is implemented by the UMPT.

Figure 3-9 Network topology for SSL applied to the OM channel of a co-MPT multimode base station



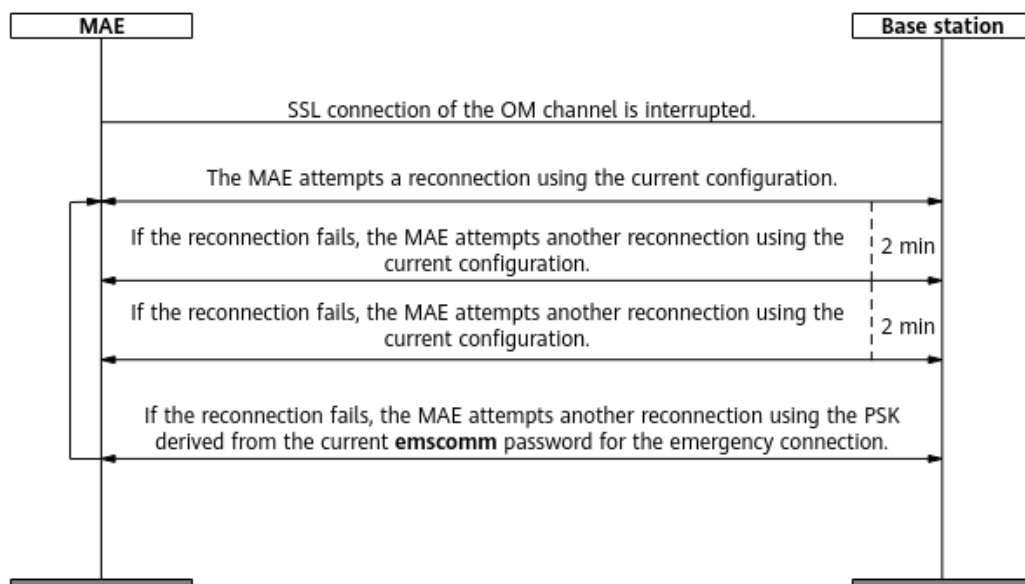
For a hybrid-MPT multimode base station, OM channels must be established between each separate-MPT main control board and the MAE, and between the co-MPT main control board and the MAE.

Emergency SSL Connection

The SSL connection of the OM channel between the base station and the MAE/NetEco may be interrupted if their OM channel SSL configurations are inconsistent (due to reasons such as certificate expiration, absence, or damage). To ensure the SSL connection reliability of the OM channel, the MAE/NetEco attempts to establish an emergency SSL connection to reconnect to the base station. [Figure](#)

3-10 uses the SSL reconnection between the MAE and base station as an example to show the detailed procedure.

Figure 3-10 SSL reconnection between the MAE and base station



1. When the SSL connection of the OM channel between the base station and MAE is interrupted, the MAE immediately attempts to reconnect to the base station using the current SSL configuration.
2. If the reconnection in **1** fails, the MAE waits for 2 minutes and attempts another reconnection using the current SSL configuration.
3. If the reconnection in **2** fails, the MAE waits for another 2 minutes and makes a third reconnection attempt using the current SSL configuration.
4. If the reconnection in **3** fails, the MAE immediately tries to reconnect using the PSK derived from the current **emscomm** password for the emergency connection. For details about PSK principles, see **3.1.2.2 SSL Connection Establishment**.
5. If the reconnection in **4** fails, the MAE returns to **1** and retries the reconnection process.

After the emergency SSL connection is successfully established, the base station reports ALM-26247 Configuration Failure with **Specific Problem** being **OMCH TLS Config Data Inconsistent**. This alarm is cleared when the TLS connection is restored and the emergency SSL connection is no longer used.

 NOTE

The emergency SSL connection is supported only by the MAE/NetEco and base station using TLS1.3. An emergency SSL connection can be established between the base station and the NetEco only after such a connection is available between the base station and the MAE. If the base station is connected only to the NetEco but not to the MAE, the NetEco cannot be connected to the base station through an emergency SSL connection.

If the configurations of the base station and MAE are normal but the network is unstable, an emergency SSL connection may be incorrectly used.

After an OM channel is set up, the base station checks the negotiation status of this channel every minute. If the SSL connection setup fails and operator's digital certificates are used for identity authentication, the base station checks the digital certificates used by the OM channel. If a digital certificate is abnormal (for example, the digital certificate has been revoked or expired, or the certificate file does not exist), an automatic certificate obtaining procedure is triggered. For details, see *PKI*.

After the SSL connection of the OM channel is interrupted, the MAE attempts to reconnect to the base station using an emergency SSL connection. To prevent the emergency SSL connection from affecting the certificate re-application mechanism of the base station, the base station rejects the MAE's first attempt (as described in 4). An emergency SSL connection may be successfully established only upon the MAE's second attempt (as described in 4). This means that an emergency SSL connection can be used to resume the OM channel 7–9 minutes after the interruption.

After the emergency SSL connection is established, the configuration interface in this version supports synchronization and delivery of the setting of the cipher suite TLS_AES_256_GCM_SHA384 for the **TLSPOLICY.CIPHERSUITE (5G gNodeB, LTE eNodeB)** parameter. This cipher suite takes effect even if it is disabled.

3.1.3.1.2 OM Channel of the BSC6900/BSC6910 (GSM/UMTS)

The OM channel between the BSC6900/BSC6910 and the MAE supports three connection modes: plaintext, plaintext+SSL, and SSL only. Matched connection modes must be configured on the BSC6900/BSC6910 and MAE sides to ensure proper serving of the OM channel.

The BSC6900/BSC6910 acts as an SSL server, and the default connection mode is SSL only. If the MAE connects to an NE in plaintext mode, the connection mode on the MAE needs to be set to the compatibility mode or SSL only mode to match the connection mode of the NE. To match the connection mode of the MAE, changing the SSL connection mode of the NE to the plaintext mode is also allowed after NE login through the Web LMT.

The SSL authentication mode of the OM channel depends on the data configurations on both the MAE and the BSC6900/BSC6910 sides, as described in [Table 3-11](#).

Table 3-11 SSL authentication mode of the OM channel of the BSC6900/BSC6910

SSL Authentication Mode	Authentication Mode on the MAE Side	Authentication Mode (SSLAUTHMODE. AUTHMODE Setting) on the BSC6900/BSC6910 Side	Deployment Requirement
Non-authentication between the BSC6900/BSC6910 and MAE	Anonymous	NONE	Both sides support the same anonymous authentication algorithm.
Unidirectional authentication of the BSC6900/BSC6910 by the MAE	Authentication of the NE by the OSS	NONE	The BSC6900/BSC6910 is preconfigured with an operator-issued device certificate and the operator's root certificate. The MAE is preconfigured with the operator's root certificate.
Bidirectional authentication between the BSC6900/BSC6910 and MAE	Bidirectional	PEER	Both the BSC6900/BSC6910 and MAE are preconfigured with an operator-issued device certificate and the operator's root certificate.
Unidirectional authentication of the MAE by the BSC6900/BSC6910	Authentication of the OSS by the NE	PEER	The BSC6900/BSC6910 is preconfigured with the operator's root certificate. The MAE is preconfigured with an operator-issued device certificate and the operator's root certificate.

For details about digital certificates of the BSC6900/BSC6910, see *PKI*.

3.1.3.1.3 OM Channel of the BSC6960 (GSM/UMTS)

The OM channel between the BSC6960 and the MAE supports three connection modes: plaintext, plaintext+SSL, and SSL only. Matched connection modes must be

configured on the BSC6960 and MAE sides to ensure proper serving of the OM channel.

The BSC6960 acts as an SSL server, and the default connection mode is SSL only. If the MAE connects to an NE in plaintext mode, the connection mode on the MAE needs to be set to the compatibility mode or SSL only mode to match the connection mode of the NE. To match the connection mode of the MAE, changing the SSL connection mode of the NE to the plaintext mode is also allowed after NE login through the Web LMT.

Before configuring SSL, set the connection type between the BSC6960 and MAE to SSL and set the authentication mode to "verify peer certificate" on the MAE. In addition, preconfigure the operator-issued device certificate and the operator's root certificate on the MAE.

The SSL authentication mode of the OM channel between the BSC6960 and MAE is determined based on the configurations on both sides, as described in [Table 3-12](#).

Table 3-12 SSL authentication mode of the OM channel between the BSC6960 and the MAE

SSL Authentication Mode	Authentication Mode on the MAE Side	Authentication Mode (SSL.AUTHMODE Setting) on the BSC6960 Side	Deployment Requirement
Non-authentication between the BSC6960 and MAE	Anonymous	NONE	None
Unidirectional authentication of the BSC6960 by the MAE	Authentication of the NE by the OSS	NONE	The BSC6960 is preconfigured with an operator-issued device certificate and the operator's root certificate. The MAE is preconfigured with the operator's root certificate.
Bidirectional authentication between the BSC6960 and MAE	Bidirectional	PEER	Both the BSC6960 and MAE are preconfigured with an operator-issued device certificate and the operator's root certificate.

SSL Authentication Mode	Authentication Mode on the MAE Side	Authentication Mode (SSL.AUTHMODE Setting) on the BSC6960 Side	Deployment Requirement
Unidirectional authentication of the MAE by the BSC6960	Authentication of the OSS by the NE	PEER	The BSC6960 is preconfigured with the operator's root certificate. The MAE is preconfigured with an operator-issued device certificate and the operator's root certificate.

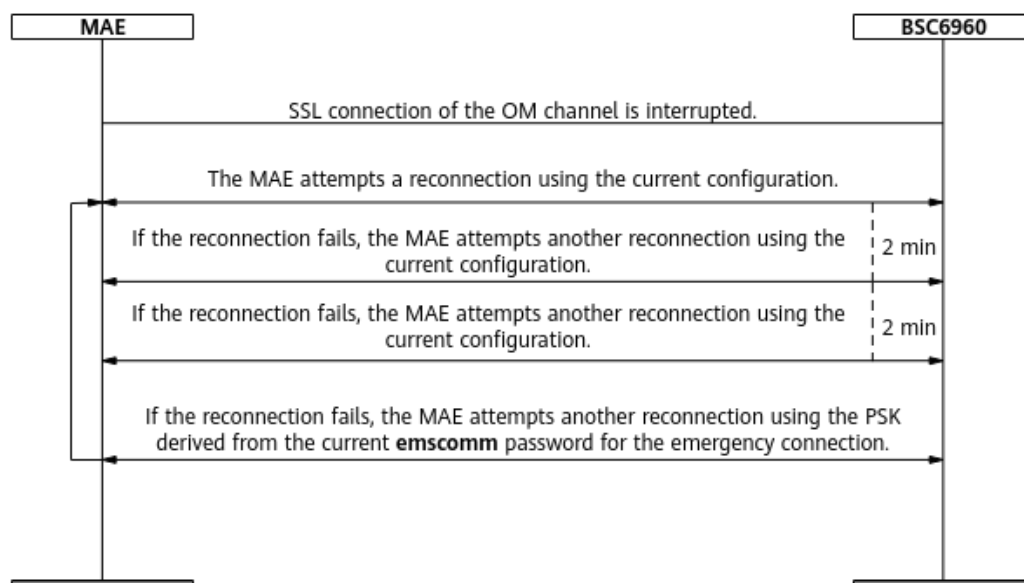
For details about digital certificates of the BSC6960, see *PKI*.

When the PKI system is deployed on an operator's network, it is recommended that the BSC6960 and the MAE use the operator's certificates to authenticate each other. In this case, set the **APPCERT.APPTYPE** parameter to **SSL** and set the **APPCERT.APPCERT** parameter to the device certificate used for SSL authentication.

When a BSC6960 works as an SSL client or server, certificates in the SSL trust group specified by the **SSL.TRUSTGROUPID** parameter can be used to verify the device certificates of the peer end. If this parameter is not specified, all trust certificates of the BSC6960 are used to verify the device certificates of the peer end by default.

The BSC6960 supports an emergency SSL connection. The SSL connection of the OM channel between the BSC6960 and the MAE may be interrupted if their OM channel SSL configurations are inconsistent (due to reasons such as certificate expiration, absence, or damage). To ensure the SSL connection reliability of the OM channel, the MAE attempts to establish an emergency SSL connection to reconnect to the BSC6960. [Figure 3-11](#) uses the SSL reconnection between the MAE and BSC6960 as an example to show the detailed procedure.

Figure 3-11 SSL reconnection between the MAE and BSC6960



1. When the SSL connection of the OM channel between the BSC6960 and MAE is interrupted, the MAE immediately attempts to reconnect to the BSC6960 using the current SSL configuration.
2. If the reconnection in **1** fails, the MAE waits for 2 minutes and attempts another reconnection using the current SSL configuration.
3. If the reconnection in **2** fails, the MAE waits for another 2 minutes and makes a third reconnection attempt using the current SSL configuration.
4. If the reconnection in **3** fails, the MAE immediately tries to reconnect using the PSK derived from the current **emscomm** password for the emergency connection. For details about PSK principles, see [3.1.2.2 SSL Connection Establishment](#).
5. If the reconnection in **4** fails, the MAE returns to **1** and retries the reconnection process.

After the emergency SSL connection is successfully established, the BSC6960 reports ALM-20984 Configuration Failure with **Specific Problem** being **OMCH TLS Config Data Inconsistent**. This alarm is cleared when the TLS connection is restored and the emergency SSL connection is no longer used.

 NOTE

The emergency SSL connection is supported only by the MAE and BSC6960 using TLS1.3. If the configurations of the MAE are normal but the network is unstable, an emergency SSL connection may be incorrectly used.

After an OM channel is set up, the BSC6960 checks the negotiation status of this channel every minute. If the SSL connection setup fails and operator's digital certificates are used for identity authentication, the BSC6960 checks the digital certificates used by the OM channel. If a digital certificate is abnormal (for example, the digital certificate has been revoked or expired, or the certificate file does not exist), an automatic certificate obtaining procedure is triggered. For details, see *PKI*.

After the SSL connection of the OM channel is interrupted, the MAE attempts to reconnect to the BSC6960 using an emergency SSL connection. To prevent the emergency SSL connection from affecting the certificate re-application mechanism of the BSC6960, the BSC6960 rejects the MAE's first attempt (as described in 4). An emergency SSL connection may be successfully established only upon the MAE's second attempt (as described in 4). This means that an emergency SSL connection can be used to resume the OM channel 7–9 minutes after the interruption.

After the emergency SSL connection is established, the configuration interface in this version supports synchronization and delivery of the setting of the cipher suite TLS_AES_256_GCM_SHA384 for the **TLSPOLICY.CIPHERSUITE** parameter. This cipher suite takes effect even if it is disabled.

3.1.3.2 FTP Transmission

The base station and base station controller support FTP over SSL (FTPS) and allow for the FTP status inspection function of the firewall.

The FTP status inspection function allows a firewall to automatically identify FTP messages and enable temporarily assigned FTP ports.

To ensure inspection of the port information about FTP data channels by a firewall, the FTP control channel must be switched to the plaintext transmission mode after account and password authentication using the FTPS protocol is complete.

FTPS is mainly applicable to file transmission between the base station and the MAE, between the base station and the base station controller, and between the base station controller and the MAE.

Table 3-13 describes the application scenarios for FTPS.

Table 3-13 Application scenarios for FTPS

FTPS Application	Description
The base station/ BSC6960 functions as the FTPS client.	• The FTPCLT.ENCYRMODE (5G gNodeB, LTE eNodeB) parameter specifies the transmission encryption mode of the base station. It is recommended that this parameter be set to Encrypted to enable FTPS. If this parameter is set to Plaintext, data will be transmitted in plaintext over FTP connections, which may cause information leakage and tampering. • The FTPCLT.SSLCERTAUTH (5G gNodeB, LTE eNodeB) parameter specifies whether to perform SSL certificate authentication on the FTP server. It is recommended that this parameter be set to Yes to authenticate the FTP server certificate. If this parameter is set to No, the base station does not authenticate the peer certificate and cannot determine whether the FTP server identity is valid. As a result, the FTP server may be spoofed. • The FTPCLT.SPTSTATEFWL (5G gNodeB, LTE eNodeB) parameter specifies whether FTPS connection setup is allowed when a stateful inspection firewall is configured. • The FTPCLT.POLICYID (5G gNodeB, LTE eNodeB) parameter specifies the TLS configuration policy of an FTPS connection. • The FTPCLT.TLSSECCSSW (5G gNodeB, LTE eNodeB) parameter specifies the TLS security algorithm switch. It is recommended that this parameter be set to ON so that only secure cipher suites are negotiated. If this parameter is set to OFF, the base station is compatible with non-AEAD and non-PFS weak security cipher suites to meet interoperability requirements. However, the encryption algorithms may be cracked. • The FTPCLT.MINDHLEN (5G gNodeB, LTE eNodeB) parameter specifies the minimum length of the Diffie-Hellman (DH) group allowed at the peer end when SSL connection is used and peer certificate authentication is required. It is recommended that this parameter be set to DH_2048 or a value indicating a larger minimum length of the DH group allowed. If the configured minimum length of the DH group is less than 2048 bits, the negotiated key is insufficiently secure and may be cracked. • The FTPCLT.TRUSTGROUPID (LTE eNodeB, 5G gNodeB) parameter specifies the SSL trust group in which the certificates are used to verify the device certificates of the peer end. If this parameter is not

FTPS Application	Description
	specified, all trust certificates of the base station are used to verify the device certificates of the peer end by default.
The BSC6910 functions as the FTPS client.	• The FTPSCLT.ENCRYMODE parameter specifies the transmission encryption mode of the base station controller. • The FTPSCLT.SSLCERTAUTH parameter specifies whether to perform SSL certificate authentication on the FTP server. • The FTPSCLT.SPTSTATEFWL parameter specifies whether FTPS connection setup is allowed when a stateful inspection firewall is configured.
The BSC6910 functions as the FTPS server.	The FTPSSRV.ENCRYMODE parameter specifies the transmission encryption mode of the base station controller. The FTPSSRV.SSLVER parameter specifies the negotiated SSL version of the base station controller.

 NOTE

The certificates used for FTPS authentication are the same as those used for SSL authentication of the OM channel.

If a connection with a third-party server or a device that is not configured with a Huawei initial certificate is required, the overall encryption mode is SSL encryption, but the peer does not support SSL encryption, the **ADD SOFTPSCLT** command needs to be executed to add the IP address of the third-party server or device for data transmission in plaintext. This ensures service availability.

When the TLS security algorithm switch is turned off, weakly secure cipher suites which do not support authenticated encryption with associated data (AEAD) or perfect forward secrecy (PFS) are allowed, posing security risks. When this switch is turned on, only the negotiated secure cipher suites are allowed for the TLS, and weakly secure cipher suites which do not support AEAD or PFS cannot be used.

FTPS uses the anonymous cipher suite, for which man-in-the-middle attacks may be launched. Therefore, the corresponding function has been removed from gNodeBs. For compatibility reasons, this function is temporarily retained for other RATs and will be disused in later versions.

3.1.3.3 HTTP Transmission

The base station and base station controller support HTTP over SSL (HTTPS). HTTPS is applicable to communications between the base station and the Web LMT, and between the base station controller and the Web LMT.

The **WEBLOGINPOLICY.POLICY** parameter specifies the login policy of the Web LMT for the base station controller.

 NOTE

The certificates used for HTTPS authentication are the same as those used for SSL authentication of the OM channel. The corresponding root certificate must be preconfigured on the Web LMT. Otherwise, when you attempt to log in to the Web LMT, a dialog box is displayed, indicating that the certificate is unreliable and asking whether to continue. If you click **Yes**, you are allowed to log in to the Web LMT.

If the **Extended Key Usage** parameter has been configured on the CA server, **Extended Key Usage** must contain **server authentication**. Otherwise, certificates issued by this CA server cannot be used to establish HTTPS connections. For details about the **Extended Key Usage** parameter, see RFC 5280 X.509 v3.

When a base station works as an HTTPS client or server, certificates in the SSL trust group specified by the **CA.TRUSTGROUPID (LTE eNodeB, 5G gNodeB)** parameter can be used to verify the device certificates of the peer end. If this parameter is not specified, all trust certificates of the base station are used to verify the device certificates of the peer end by default.

The **SSLVER (5G gNodeB, LTE eNodeB)** parameter specifies the negotiated SSL version of the Web LMT for the base station/base station controller. In earlier versions, the SSL version of the Web LMT is the same as that of the OM channel. After an upgrade to this version, the SSL version of the Web LMT inherits that of the OM channel.

 NOTE

The Web LMT supports TLS1.2 and TLS1.3 by default. When setting **SSLVER (5G gNodeB, LTE eNodeB)** for the Web LMT, ensure that the browser is configured with the corresponding SSL version. Otherwise, the Web LMT cannot be opened using the browser.

To prevent attacks and ensure security, it is recommended that TLS1.2 and TLS1.3 be used.

HTTPS is also applicable to periodic CRL acquisition and the certificate management protocol v2 (CMPv2) information exchange between the base station/base station controller and CA server. In this case, the base station/base station controller works as the HTTPS client.

 NOTE

In CMPv2 scenarios, the TLS security algorithm switch is specified by the **CA.TLSSECCSSW (5G gNodeB, LTE eNodeB)** parameter. When this switch is turned off, weakly secure cipher suites which do not support AEAD or PFS are allowed, posing security risks. When this switch is turned on, only the negotiated secure cipher suites are allowed for the TLS, and weakly secure cipher suites which do not support AEAD or PFS cannot be used.

For periodic CRL acquisition, the preceding weakly secure cipher suites are incompatible and non-peer-authentication mode is not supported.

3.1.3.4 LDAP Transmission

The base station/base station controller supports the application of SSL to LDAP, that is, StartTLS for LDAP. If the access method and connection mode in the **ADD CRLTSK** command are set to **LDAP** and **SSL**, respectively, StartTLS for LDAP will be applied to the base station, base station controller, and LDAP server for TLS connection establishment. After a successful TLS handshake, all LDAP packets are transmitted on secure links.

Table 3-14 describes the application of StartTLS for LDAP.

Table 3-14 Application of StartTLS for LDAP

Application of StartTLS for LDAP	Description
The base station/BSC6960 functions as the LDAP client.	- The CRLTSK.CONNMODE (5G gNodeB, LTE eNodeB) parameter specifies the transmission encryption mode of the base station. It is recommended that this parameter be set to SSL to enable StartTLS for LDAP. If this parameter is set to PLAINTEXT, data will be transmitted in plaintext over LDAP connections, which may cause information leakage and tampering. - The CRLTSK.MINDHLEN (5G gNodeB, LTE eNodeB) parameter specifies the minimum length of the Diffie-Hellman (DH) group allowed at the peer end when SSL connection is used and peer certificate authentication is required. It is recommended that this parameter be set to DH_2048 or a value indicating a larger minimum length of the DH group allowed. If the configured minimum length of the DH group is less than 2048 bits, the negotiated key is insufficiently secure and may be cracked. NOTE To prevent incorrect configurations, you can run the SET SECPLY command to adjust the configuration security level. The DH_0, DH_512, and DH_1024 values are not allowed to be used. For details, see <i>OM Security</i>. - The CRLTSK.AUTHPEER (5G gNodeB, LTE eNodeB) parameter specifies whether to authenticate the certificate of the peer end. It is recommended that this parameter be set to ENABLE to authenticate the LDAP server certificate. If this parameter is set to DISABLE, the base station does not authenticate the peer certificate and cannot determine whether the LDAP server identity is valid. As a result, the LDAP server may be spoofed.

Application of StartTLS for LDAP	Description
	NOTE To prevent incorrect configurations, you can run the SET SECPY command to adjust the configuration security level. The CRLTSK.AUTHPEER (5G gNodeB, LTE eNodeB) parameter is not allowed to be set. For details, see <i>OM Security</i>. • The CRLTSK.POLICYID (5G gNodeB, LTE eNodeB) parameter specifies the TLS configuration policy of an LDAP connection. • The CRLTSK.TLSSECCSSW (5G gNodeB, LTE eNodeB) parameter specifies the TLS security algorithm switch. It is recommended that this parameter be set to ON so that only secure cipher suites are negotiated. If this parameter is set to OFF, the base station is compatible with non-AEAD and non-PFS weak security cipher suites to meet interoperability requirements. However, the encryption algorithms may be cracked. NOTE To prevent incorrect configurations, you can run the SET SECPY command to adjust the configuration security level. The CRLTSK.TLSSECCSSW (5G gNodeB, LTE eNodeB) parameter is not allowed to be set. For details, see <i>OM Security</i>. • The CRLTSK.TRUSTGROUPID (LTE eNodeB, 5G gNodeB) parameter specifies the SSL trust group in which the certificates are used to verify the device certificates of the peer end. If this parameter is not specified, all trust certificates of the base station are used to verify the device certificates of the peer end by default. • The CRLTSK.CERTNAME (LTE eNodeB, 5G gNodeB) parameter specifies whether the device certificate configured in CERTMK MO is used for authentication when a TLS connection is established with the LDAP server. If this parameter is

Application of StartTLS for LDAP	Description
	not specified, the system uses Huawei initial certificates. NOTE For details about the device certificate configured in the CERTMK MO, see <i>PKI</i> .
The base station controller functions as the LDAP client.	The CRLTSK.AUTHPEER parameter specifies whether to authenticate the identity of the peer end.

 NOTE

When the TLS security algorithm switch is turned off, weakly secure cipher suites which do not support AEAD or PFS are allowed, posing security risks. When this switch is turned on, only the negotiated secure cipher suites are allowed for the TLS, and weakly secure cipher suites which do not support AEAD or PFS cannot be used.

3.1.3.5 eCPRI Interface Management Plane

The base station supports eCPRI interface management plane over SSL. To enable SSL connection establishment, run the **SET RRUSECPOLICY** command on the BBU side with the **SSL.RRUTLSCONNTYPE (LTE eNodeB, 5G gNodeB)** parameter set to **TLS** or **TLS_PLAINTEXT**. After an SSL connection is established, all packets on the management plane must be transmitted over secure connections. In this scenario, the BBU functions as the SSL server, and the AAU functions as the SSL client.

If the Huawei initial certificate of an AAU is damaged, the **SET RRUSECPOLICY** command can be executed to change the authentication mode related to the BBU to **NONE**. The **LST SSLCONF** command can be executed to check whether anonymous authentication is used. The **DSP RRUTLSERRINFO** command can be executed to query the negotiated authentication mode.

 NOTE

In eCPRI scenarios, the BBU and AAU support TLS1.2 and TLS1.3. The TLS cipher suites and key update period used by the BBU and AAU in eCPRI scenarios retain factory settings and are not configurable.

When SSL connections are used, run the **SET RRUSECPOLICY** command with the **SSL.RRUAUTHMODE (LTE eNodeB, 5G gNodeB)** parameter set to **PEER** to authenticate the AAUs. If this parameter is set to **NONE**, man-in-the-middle attacks may occur.

3.2 Network Analysis

3.2.1 Benefits

SSL ensures secure communication between the client and the server by establishing an SSL connection. SSL provides the following security functions:

- Confidentiality: SSL encrypts data transmitted between communicating parties to prevent eavesdropping.
- Authentication: The communicating parties must authenticate each other before establishing an SSL connection.
- Integrity: SSL provides integrity protection for data transmitted between the communicating parties to prevent data tampering during transmission.

3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.1.3 Function Impacts](#), [3.3.2.2.3 Function Impacts](#), [3.3.2.3.3 Function Impacts](#), [3.3.2.4.3 Function Impacts](#), [3.3.2.5.3 Function Impacts](#), [3.3.2.6.3 Function Impacts](#), and [3.3.2.7.3 Function Impacts](#).

When SSL provides encryption and integrity protection, the network bandwidth utilization decreases slightly.

3.3 Requirements

3.3.1 Licenses

There are no license requirements for basic functions.

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 GBFD-113522 Encrypted Network Management

3.3.2.1.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
GSM	BTS Supporting PKI	None	<i>PKI</i>	This feature is required if digital certificates need to be used for SSL authentication.

3.3.2.1.2 Mutually Exclusive Functions

None

3.3.2.1.3 Function Impacts

None

3.3.2.2 SSL on the NodeB Side

3.3.2.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
UMTS	NodeB PKI Support	None	<i>PKI</i>	This feature is required if digital certificates need to be used for SSL authentication.

3.3.2.2.2 Mutually Exclusive Functions

None

3.3.2.2.3 Function Impacts

None

3.3.2.3 LBFD-004003 Security Socket Layer

3.3.2.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Public key infrastructure (PKI)	None	<i>PKI</i>	This feature is required if digital certificates need to be used for SSL authentication.

3.3.2.3.2 Mutually Exclusive Functions

None

3.3.2.3.3 Function Impacts

None

3.3.2.4 TDLBFD-004003 Security Socket Layer

3.3.2.4.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Public key infrastructure (PKI)	None	<i>PKI</i>	This feature is required if digital certificates need to be used for SSL authentication.

3.3.2.4.2 Mutually Exclusive Functions

None

3.3.2.4.3 Function Impacts

None

3.3.2.5 MLBFD-12000403 Security Socket Layer

3.3.2.5.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
NB-IoT	Public key infrastructure (PKI)	None	<i>PKI</i>	This feature is required if digital certificates need to be used for SSL authentication.

3.3.2.5.2 Mutually Exclusive Functions

None

3.3.2.5.3 Function Impacts

None

3.3.2.6 SSL on the Base Station Controller Side

3.3.2.6.1 Prerequisite Functions

None

3.3.2.6.2 Mutually Exclusive Functions

None

3.3.2.6.3 Function Impacts

None

3.3.2.7 FBFD-010023 Security Mechanism (SSL)

3.3.2.7.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Security Mechanism (PKI)	None	<i>PKI</i>	This feature is required if digital certificates need to be used for SSL authentication.

3.3.2.7.2 Mutually Exclusive Functions

None

3.3.2.7.3 Function Impacts

None

3.3.3 Hardware

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations
UMTS	3900 and 5900 series base stations - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

RAT	Base Station Model
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Table 3-15 describes the hardware required for deploying SSL on the eGBTS/NodeB/eNodeB/gNodeB.

Table 3-15 Hardware required for deploying SSL on the eGBTS/NodeB/eNodeB/gNodeB

NE	Board Configuration	Board That Provides a Port for Connecting to the Transport Network	Port Type
eGBTS	UMPT/UMDU/MDUC	UMPT/UMDU/MDUC	Ethernet port
eGBTS	UMPT+UTRPc	UTRPc	Ethernet port
NodeB	UMPT/UMDU/MDUC	UMPT/UMDU/MDUC	Ethernet port
NodeB	UMPT+UTRPc	UTRPc	Ethernet port
eNodeB	UMPT/UMDU	UMPT/UMDU	Ethernet port
eNodeB	UMPT+UTRPc	UTRPc	Ethernet port
gNodeB	UMPT	UMPT	Ethernet port

RF Modules

No requirements

3.3.4 Others

For the Base Station

If an operator-issued device certificate is used for SSL authentication, the PKI system needs to be deployed on the network and the PKI feature needs to be activated on the base station side. For details about how to deploy the PKI system, see *PKI*.

For the Base Station Controller (GSM/UMTS)

If digital certificates are required to authenticate the SSL connection of the OM channel, ensure that the certificates have been preconfigured on the OMU board

of the BSC6900/BSC6910 and the CMPU board of the BSC6960. For details about how to configure digital certificates for the base station controller, see *PKI*.

3.4 Operation and Maintenance

3.4.1 For the Base Station

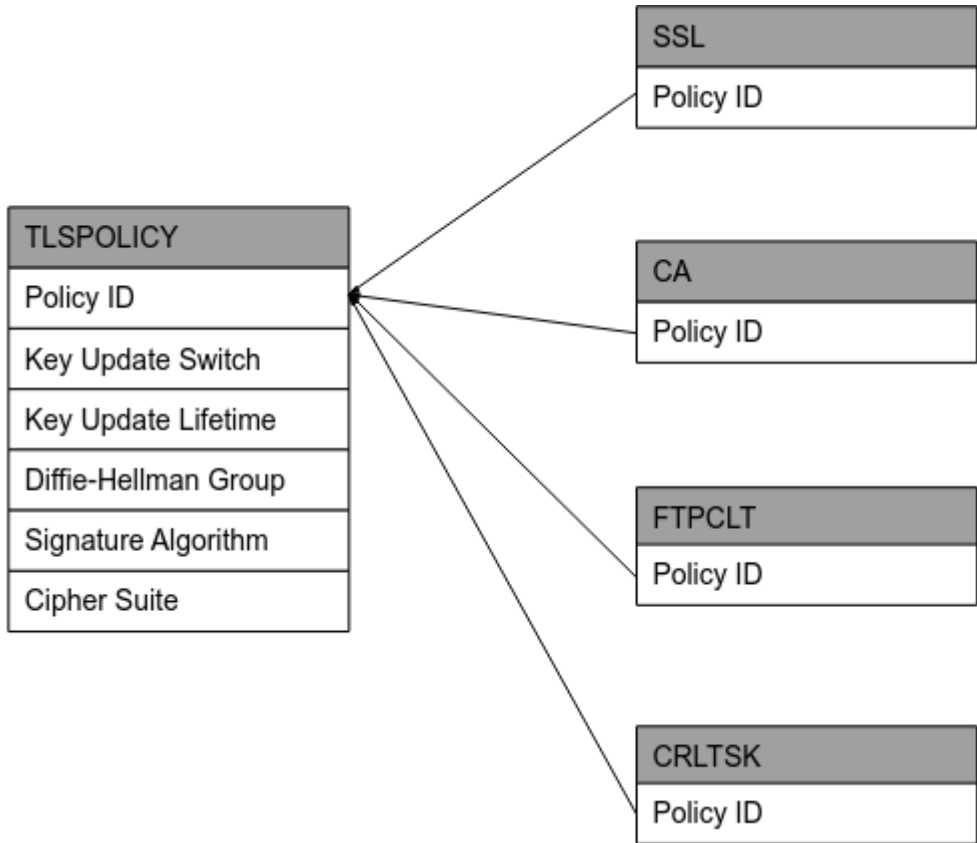
3.4.1.1 When to Use

The public IP network cannot ensure transmission security. Therefore, when operators use the public IP network to carry wireless services, it is recommended that SSL be used to provide transmission security for the OM channel. For networks with high OM channel reliability requirements, SSL PSK is recommended to prevent interruption of services on the OM channel caused by certificate damage.

3.4.1.2 Data Configuration

According to RFC 8446, TLS1.3 differs greatly from TLS1.2. Therefore, a new MO **TLSPOLICY** is introduced to configure TLS1.3 key update and encryption algorithms. Users are allowed to configure one or more TLS configuration policies based on the actual network conditions. One TLS configuration policy can be bound to each SSL application scenario (OM channel/FTP/CMPv2/LDAP). [Figure 3-12](#) shows the TLS1.3 configuration policy.

Figure 3-12 TLS1.3 configuration policy



As shown in [Figure 3-12](#):

- The **SSL** and **TLSPOLICY** MOs use **POLICYID** to specify the TLS configuration policy used when SSL is applied to the OM channel.
- The **CA** and **TLSPOLICY** MOs use **POLICYID** to specify the TLS configuration policy used when SSL is applied to CMPv2-based interaction scenarios.
- The **FTPCLT** and **TLSPOLICY** MOs use **POLICYID** to specify the TLS configuration policy used when SSL is applied to FTP applications.
- The **CRLTSK** and **TLSPOLICY** MOs use **POLICYID** to specify the TLS configuration policy used when SSL is applied to LDAP/HTTP applications.

3.4.1.2.1 Data Preparation

The SSL configuration data is the same for the eGBTS/NodeB/eNodeB/gNodeB. This section describes only the SSL configuration. For the configuration of the PKI feature, see *PKI*.

SSL Connection for the OM Channel

(Optional) Collect the connection type supported by the base station and USU (the corresponding MML command is **SET CONNTYPE**), as described in [Table 3-16](#).

Table 3-16 Connection type

Parameter Name	Parameter ID	Setting Notes
Connection Type	SSL.CONNTYPE (LTE eNodeB, 5G gNodeB)	The default value is SSL . It is recommended that this parameter be set to SSL to enable SSL-based security connection. If this parameter is set to ALL , all connection types are supported. In this case, the base station can be connected in non-SSL mode, and information transmitted in plaintext may be disclosed or tampered with.

Prepare data about the SSL authentication mode of the OM channel (the corresponding MML command is **SET SSLAUTHMODE**), as described in [Table 3-17](#).

Table 3-17 SSL authentication mode

Parameter Name	Parameter ID	Setting Notes
Authentication Mode	SSL.AUTHMODE (LTE eNodeB, 5G gNodeB)	The default value of this parameter is PEER . Ensure that certificates are correctly configured on the MAE and base station. It is recommended that this parameter be set to PEER to verify the MAE identity. If this parameter is set to NONE , the MAE identity cannot be determined. In this case, the MAE may be spoofed.

(Optional) Prepare data about the SSL connection version of the OM channel (the corresponding MML command is **SET SSLCONF**), as described in [Table 3-18](#).

Table 3-18 SSL connection version

Parameter Name	Parameter ID	Setting Notes
TLS/SSL Version	SSL.VERSION (5G gNodeB, LTE eNodeB)	TLS1.2 and TLS1.3 are enabled by default.

(Optional) Prepare certificate data, as described in [Table 3-19](#). The parameters in the following MO specify the device certificate used for SSL authentication.

Table 3-19 Certificate configuration

Parameter Name	Parameter ID	Setting Notes
Application Type	APPCERT.APPTYPE (5G gNodeB, LTE eNodeB)	Set this parameter to SSL .
Certificate File Name	APPCERT.APPCERT (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan. If an operator-issued device certificate is used for authentication, set this parameter to the name of the certificate.

(Optional) Prepare information about the trust group, as described in [Table 3-20](#). The trust group is used for peer verification in SSL authentication.

Table 3-20 Information about the trust group

Parameter Name	Parameter ID	Setting Notes
Trust Group ID	<i>SSL.TRUSTGROUPID (LTE eNodeB, 5G gNodeB)</i>	This parameter is used for verifying the device certificates of the peer end. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the TRUSTGROUP MO are used to authenticate the peer end. If this parameter is set to 65535 , no trust group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.

(Optional) Configure the TLS configuration policy for the OM channel, as described in [Table 3-21](#).

Table 3-21 TLS configuration policy for the OM channel

Parameter Name	Parameter ID	Setting Notes
Policy ID	<i>SSL.POLICYID (5G gNodeB, LTE eNodeB)</i>	The default value of this parameter is 0 . For the TLS configuration policy numbered 0, the key update function is enabled, the key update period is 60 minutes, and all TLS1.3 cryptographic algorithms are enabled. Before setting this parameter, ensure that the corresponding TLS configuration policy has been configured. If the corresponding TLS configuration policy does not exist, configure a TLSPOLICY MO and then set this parameter to the corresponding policy ID.

(Optional) Configure SSL PSK for the OM channel, as described in [Table 3-22](#).

Table 3-22 SSL PSK configuration for the OM channel

Parameter Name	Parameter ID	Setting Notes
Authentication Method	<i>SSL.AUTHMETH (LTE eNodeB, 5G gNodeB)</i>	It is recommended that this parameter be set to CERT_PRIORITY for networks that have high reliability requirements for the OM channel. In addition, ensure that both certificate-based authentication and PSK-based authentication are supported.
PSK ID	<i>SSL.PID (LTE eNodeB, 5G gNodeB)</i>	The value can contain 1 to 127 characters and must be the same as that on the MAE.
PSK KEY MATERIAL	<i>SSL.PKEY (LTE eNodeB, 5G gNodeB)</i>	The value of this parameter must contain at least 20 characters, meeting the complexity requirements for uppercase letters, lowercase letters, digits, and special characters. The value must be the same as that on the MAE.

 NOTE

Before activating the SSL feature on a separate-MPT multimode base station, configure SSL data for each mode separately.

Before activating the SSL feature on a co-MPT multimode base station, configure only a set of SSL data, which is shared by different modes of the base station.

The SSL PSK configuration for the OM channel can be set or modified on both the base station and MAE sides. It is recommended that the configuration be set or modified on the MAE and then delivered to the base station to ensure that the configuration on the base station is consistent with that on the MAE.

FTP Client Connection

Prepare data about the FTP client connection, as described in [Table 3-23](#).

Table 3-23 FTP client connection data

Parameter Name	Parameter ID	Setting Notes
Transport Encrypted Mode	FTPCLT.ENCYRMODE <i>(5G gNodeB, LTE eNodeB)</i>	It is recommended that this parameter be set to Encrypted to enable FTPS. If this parameter is set to Plaintext , data will be transmitted in plaintext over FTP connections, which may cause information leakage and tampering.
Support State Firewall	FTPCLT.SPTSTATEFW <i>L (5G gNodeB, LTE eNodeB)</i>	Set this parameter based on whether the stateful firewall is deployed on the customer's live network.
Support SSL Certificate Authentication	FTPCLT.SSLCERTAUTH <i>(5G gNodeB, LTE eNodeB)</i>	It is recommended that this parameter be set to Yes to authenticate the FTP server certificate. If this parameter is set to No, the peer certificate is not authenticated and whether the FTP server identity is valid cannot be determined. As a result, the FTP server may be spoofed. In this version, the setting of this parameter can still be synchronized and delivered through the interface, and the setting can still take effect. In later versions, however, this parameter will be disused. Therefore, avoid using this parameter. To prevent incorrect configurations, you can run the SET SECPLY command to adjust the configuration security level. The No value is not allowed to be used.
Policy ID	FTPCLT.POLICYID <i>(5G gNodeB, LTE eNodeB)</i>	The default value of this parameter is 0 . For the TLS configuration policy numbered 0, the key update function is enabled, the key update period is 60 minutes, and all TLS1.3 cryptographic algorithms are enabled. Before setting this parameter, ensure that the corresponding TLS configuration policy has been configured. If the corresponding TLS configuration policy does not exist, configure a TLSPOLICY MO and then set this parameter to the corresponding policy ID.

Parameter Name	Parameter ID	Setting Notes
TLS Secure Cipher Suite Switch	FTPCLT.TLSSECCSSW <i>(5G gNodeB, LTE eNodeB)</i>	The default value of this parameter is ON , indicating that only secure cipher suites are negotiated. If this parameter is set to OFF , the base station is compatible with non-AEAD and non-PFS weak security cipher suites to meet interoperability requirements. However, the encryption algorithms may be cracked. To prevent incorrect configurations, you can run the SET SECPY command to adjust the configuration security level. The OFF value is not allowed to be used.
Trust Group ID	FTPCLT.TRUSTGROUP ID <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter is used for verifying the device certificates of the peer end. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the TRUSTGROUP MO are used to authenticate the peer end. If this parameter is set to 65535 , no trust group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.

Policy for a Web LMT Login

Prepare data about the policy for Web LMT login (the corresponding MML command is **SET WEBLOGINPOLICY**), as described in [Table 3-24](#).

Table 3-24 Policy for a Web LMT login

Parameter Name	Parameter ID	Setting Notes
TLS/SSL Version	WEBLMT.SSLVER <i>(5G gNodeB, LTE eNodeB)</i>	TLS1.2 and TLS1.3 are enabled by default. If a third-party browser is used to open the Web LMT, ensure that the browser is enabled with TLS1.2 or TLS1.3.

SSL Renegotiation Management

Prepare data about SSL renegotiation management (the corresponding MML command is **SET SSLCONF**), as described in [Table 3-25](#).

Table 3-25 SSL renegotiation management

Parameter Name	Parameter ID	Setting Notes
Renegotiation Switch	SSL.RENEGO (5G gNodeB, LTE eNodeB)	The recommended value of this parameter is ENABLE .
Renegotiation Interval	SSL.RENEGOINTERVAL (5G gNodeB, LTE eNodeB)	The recommended value of this parameter is 60 .

NOTE

The SSL renegotiation management configuration does not take effect for TLS1.3.

SSL Key Update Management

Prepare data about SSL key update management, as described in [Table 3-26](#).

Table 3-26 SSL key update management

Parameter Name	Parameter ID	Setting Notes
POLICY ID	TLSPOLICY.POLICYID (5G gNodeB, LTE eNodeB)	This parameter is used together with: • SSL.POLICYID (5G gNodeB, LTE eNodeB) to specify the TLS configuration policy used when SSL is applied to the OM channel. • CA.POLICYID (5G gNodeB, LTE eNodeB) to specify the TLS configuration policy used when SSL is applied to CMPv2-based interaction scenarios. • FTPCLT.POLICYID (5G gNodeB, LTE eNodeB) to specify the TLS configuration policy used when SSL is applied to FTP applications. • CRLTSK.POLICYID (5G gNodeB, LTE eNodeB) to specify the TLS configuration policy used when SSL is applied to LDAP applications.

Parameter Name	Parameter ID	Setting Notes
Key Update Switch	TLSPOLICY.KEYUPDA TESW (5G gNodeB, LTE eNodeB)	The default value ON is recommended. If this parameter is set to OFF , the key may not be updated for a long time and may be cracked.
Key Update Lifetime	TLSPOLICY.KEYUPDA TELT (5G gNodeB, LTE eNodeB)	The default value is recommended.

SSL Cipher Suite Configuration

Prepare data about TLS1.2 cipher suites, as described in [Table 3-27](#).

Table 3-27 SSL cipher suite configuration

Parameter Name	Parameter ID	Setting Notes
Lowest Cipher Suite Level	SSL.LOWESTCSLEVEL (5G gNodeB, LTE eNodeB)	It is recommended that this parameter be set to the same value for all NEs in an operator's network. It is recommended that this parameter be set to HIGH to use an algorithm with a high security level. If this parameter is set to MEDIUM or LOW , some weak encryption algorithms are compatible, which may be cracked. It is recommended that this parameter be set to HIGH for newly deployed NEs.
Cipher Suite Name	SSLCS.CSNAME (5G gNodeB, LTE eNodeB)	This parameter specifies the name of a cipher suite, which complies with RFC 5246. Cipher suite names cannot be customized and must be defined before product delivery, as listed in Table 3-1 .

Parameter Name	Parameter ID	Setting Notes
Cipher Suite Level	SSLCS.CSLEVEL (5G gNodeB, LTE eNodeB)	This parameter specifies the security level of a cipher suite. It is recommended that this parameter retain its default value, as listed in Table 3-1. This parameter needs to be configured based on the actual situations. For example: - Set this parameter to DISABLE to disable a cipher suite. - If a cipher suite becomes a weak algorithm due to its security vulnerabilities, change its security level to LOW. For details, see Table 3-2. If the value of this parameter is changed from HIGH to MEDIUM, LOW, or DISABLE, both sides may fail to negotiate a cipher suite and consequently connections cannot be established. Therefore, exercise caution when setting this parameter.

[Table 3-28](#) describes the parameters that must be set in the **TLSPOLICY** MO to configure TLS1.3 cipher suites.

Table 3-28 TLS1.3 cipher suite configuration

Parameter Name	Parameter ID	Setting Notes
Diffie-Hellman Group	TLSPOLICY.DHGRP (5G gNodeB, LTE eNodeB)	By default, all algorithms are enabled. The algorithm SECP256R1 cannot be disabled. Set this parameter based on the network plan.
Signature Algorithm	TLSPOLICY.SIGALG (5G gNodeB, LTE eNodeB)	By default, all algorithms are enabled. The algorithm RSA_PSS_RSAE_SHA256 cannot be disabled. Set this parameter based on the network plan.
Cipher Suite	TLSPOLICY.CIPHERSUITE (5G gNodeB, LTE eNodeB)	By default, all algorithms are enabled. The algorithm TLS_AES_128_GCM_SHA256 cannot be disabled. Set this parameter based on the network plan.

Lists of Cipher Suites Supported by the Base Station Functioning as the Client

- FTP over SSL

A base station supports TLS1.2 and TLS1.3 when it functions as the FTPS client. The following table lists the SSL cipher suites supported by the base station.

No.	SSL Cipher Suite
1	ECDHE_ECDSA_AES_256_GCM_SHA384
2	ECDHE_RSA_AES_256_GCM_SHA384
3	ECDHE_ECDSA_AES_128_GCM_SHA256
4	ECDHE_RSA_AES_128_GCM_SHA256
5	DHE_RSA_AES_256_GCM_SHA384
6	DHE_RSA_AES_128_GCM_SHA256
7	RSA_AES_256_GCM_SHA384
8	RSA_AES_128_GCM_SHA256
9	RSA_AES_256_CBC_SHA256
10	DHE_RSA_AES_256_CBC_SHA256
11	RSA_AES_256_CBC_SHA
12	DHE_RSA_AES_256_CBC_SHA
13	RSA_AES_128_CBC_SHA256
14	DHE_RSA_AES_128_CBC_SHA256
15	RSA_AES_128_CBC_SHA
16	DHE_RSA_AES_128_CBC_SHA
17	DH_anon_AES_256_CBC_SHA256
18	DH_anon_AES_256_CBC_SHA
19	DH_anon_AES_128_CBC_SHA
20	DH_anon_AES_128_CBC_SHA256

- HTTP over SSL

A base station supports TLS1.2 and TLS1.3 when it functions as the CMPv2 client. The following table lists the SSL cipher suites supported by the base station.

No.	SSL Cipher Suite
1	ECDHE_ECDSA_AES_256_GCM_SHA384

No.	SSL Cipher Suite
2	ECDHE_RSA_AES_256_GCM_SHA384
3	ECDHE_ECDSA_AES_128_GCM_SHA256
4	ECDHE_RSA_AES_128_GCM_SHA256
5	DHE_RSA_AES_256_GCM_SHA384
6	DHE_RSA_AES_128_GCM_SHA256
7	RSA_AES_128_GCM_SHA256
8	RSA_AES_128_CBC_SHA
9	RSA_AES_128_CBC_SHA256
10	RSA_AES_256_GCM_SHA384
11	RSA_AES_256_CBC_SHA
12	RSA_AES_256_CBC_SHA256

- LDAP over SSL

The following table lists the cipher suites supported when StartTLS for LDAP is applied and authenticating the certificate of the peer end is required.

No.	Cipher Suite with Minimum DH Length Being DH_0	Cipher Suite with Minimum DH Length Not Being DH_0
1	ECDHE_ECDSA_AES_256_GCM_SHA384	ECDHE_ECDSA_AES_256_GCM_SHA384
2	ECDHE_RSA_AES_256_GCM_SHA384	ECDHE_RSA_AES_256_GCM_SHA384
3	ECDHE_ECDSA_AES_128_GCM_SHA256	ECDHE_ECDSA_AES_128_GCM_SHA256
4	ECDHE_RSA_AES_128_GCM_SHA256	ECDHE_RSA_AES_128_GCM_SHA256
5	DHE_RSA_AES_256_GCM_SHA384	DHE_RSA_AES_256_GCM_SHA384
6	DHE_RSA_AES_128_GCM_SHA256	DHE_RSA_AES_128_GCM_SHA256
7	DHE_RSA_AES_128_CBC_SHA	RSA_AES_128_GCM_SHA256
8	DHE_RSA_AES_128_CBC_SHA256	RSA_AES_256_CBC_SHA
9	DHE_RSA_AES_256_CBC_SHA	RSA_AES_128_CBC_SHA256
10	RSA_AES_128_GCM_SHA256	RSA_AES_256_GCM_SHA384

No.	Cipher Suite with Minimum DH Length Being DH_0	Cipher Suite with Minimum DH Length Not Being DH_0
11	RSA_AES_128_CBC_SHA	RSA_AES_256_CBC_SHA
12	RSA_AES_128_CBC_SHA256	RSA_AES_256_CBC_SHA256
13	RSA_AES_256_GCM_SHA384	None
14	RSA_AES_256_CBC_SHA	None
15	RSA_AES_256_CBC_SHA256	None

3.4.1.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Configuring SSL for the OM channel
//Configuring the type of device certificate used for SSL authentication
MOD APPCERT: APPTYPE=SSL, APPCERT="appcert.pem";
//Setting the connection policy for the FTP client
SET FTPSCLT: ENCRYPMODE=Encrypted, SPTSTATEFWL=Yes, SSLCERTAUTH=Yes, TRUSTGROUPID=65535;
//Setting the login policy of the Web LMT
SET WEBLOGINPOLICY: SSLVER=TLSV12-1&TLSV13-1;
//Setting the SSL version for the OM channel
SET SSLCONF: VERSION=TLSV12-1&TLSV13-1;
//Setting the TLS configuration policy for the OM channel
SET SSLCONF: POLICYID=0;
//Setting a trust group for the OM channel
SET SSLCONF: TRUSTGROUPID=65535;
//Configuring SSL renegotiation management
SET SSLCONF: RENEGO=ENABLE, RENEGOINTERVAL=60;
//Configuring key update management
MOD TLSPOLICY: POLICYID=0, KEYUPDATESW=ON, KEYUPDATELT=60;
//Configuring the selection policy of cipher suites
SET SSLCONF:LOWESTCSLEVEL=HIGH;
//Setting the security levels of cipher suites (The security levels of cipher suites are set based on site requirements. The following two scenarios are used as examples.)
```

- **Scenario 1:**
//When the cipher suite ECDHE_ECDSA_AES_256_GCM_SHA384 must be disabled for an operator
SET SSLCS:CSNAME=ECDHE_ECDSA_AES_256_GCM_SHA384, CSLEVEL=DISABLE;
- **Scenario 2:**
//When the cipher suite ECDHE_ECDSA_AES_256_GCM_SHA384 becomes a weak algorithm due to increasing computing capabilities
SET SSLCS:CSNAME=ECDHE_ECDSA_AES_256_GCM_SHA384, CSLEVEL=LOW;

3.4.1.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

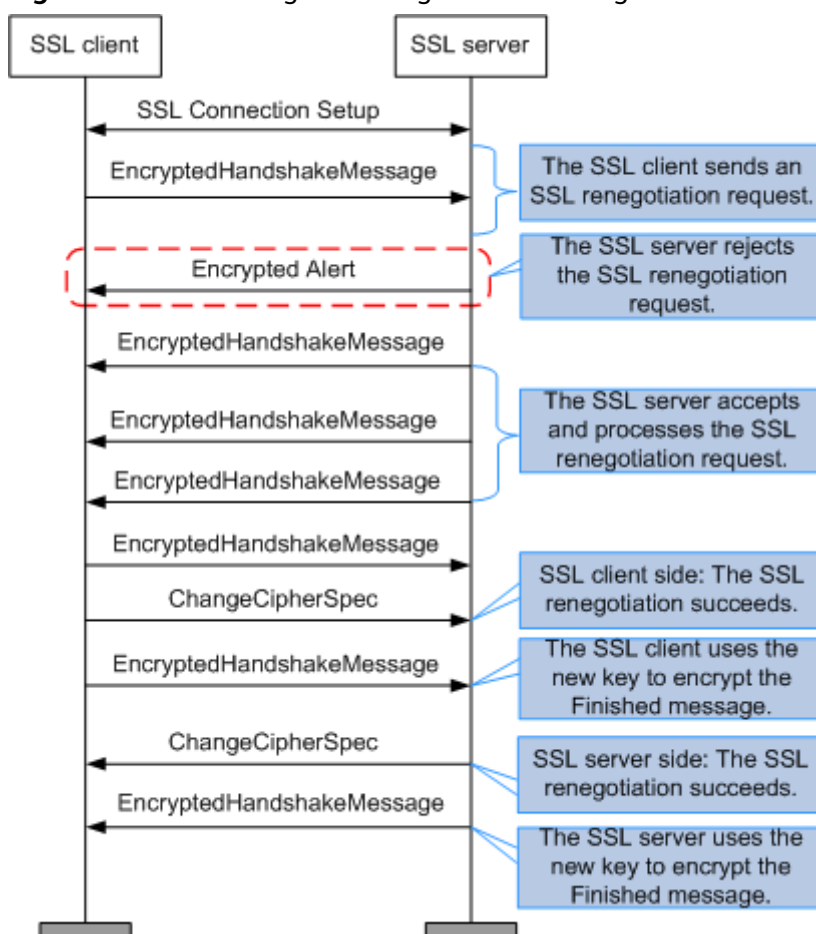
 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.4.1.3 Activation Verification

- **SSL for the OM channel**
In the SSL connection management window of the MAE, check whether the connection between the NE and the MAE is normal. If the connection is normal, SSL has been successfully activated on the OM channel.
- **FTPS**
Check whether log files are being transmitted between the base station and the MAE based on FTPS as expected. If log file transmission is normal, an FTPS connection has been successfully established between the base station and the MAE.
- **Connection between the base station and the Web LMT**
TLS1.2 and TLS1.3 are configured for the Web LMT by default. Log in to the base station through the Web LMT. A successful login to the base station indicates that a connection has been successfully configured between the base station and the Web LMT.
- **SSL renegotiation management**
Method 1:
SSL renegotiation takes place between the SSL server and the SSL client. Therefore, a method for checking whether the SSL renegotiation management function works properly is to capture packets transmitted between the SSL server and the SSL client. The following uses the base station working as the SSL server as an example to describe how to observe SSL renegotiation management.
Prerequisites:
An SSL connection has been established between the SSL server and the SSL client. Then, enable the SSL client to send an SSL renegotiation request message to the SSL server.

Figure 3-13 Observing SSL renegotiation management



Analyze the messages transmitted between the SSL server and SSL client shown in **Figure 3-13**:

- If the SSL client sends an EncryptedHandshakeMessage to the SSL server immediately after an SSL connection is established, this message is an encrypted Client Hello handshake message, indicating that the SSL client initiates an SSL renegotiation procedure.
- Upon receipt of the EncryptedHandshakeMessage from the SSL client, if the SSL server consecutively sends two to four EncryptedHandshakeMessage messages to the SSL client, the SSL server has accepted and processed the SSL renegotiation request. Among these message interactions, if both the SSL server and the SSL client send a ChangeCipherSpec message to the peer end, the SSL renegotiation procedure is completed, and the new cipher keys are used for communication.
- Upon receipt of the EncryptedHandshakeMessage from the SSL client, if the SSL server responds with an Encrypted Alert message, the SSL server has rejected the SSL renegotiation request.

Method 2:

Run the MML command **DSP SSLRENEGO** to check the value of the **Response Type** parameter. Based on the value of this parameter, the performance of the SSL renegotiation management function in reducing resource consumption and preventing DoS attacks will be evaluated.

The values of **Response Type** include **SUCCESS**, **DENY**, and **FAILURE**.

- **SUCCESS** indicates that the SSL server has accepted and processed the SSL client's renegotiation request and the SSL renegotiation is successful.
- **DENY** indicates that the SSL renegotiation management function is enabled at the SSL server and the SSL server has rejected the SSL client's renegotiation request.
- **FAILURE** indicates that the SSL server fails to complete the SSL renegotiation procedure due to exceptions not caused by the SSL renegotiation management function.

Table 3-29 describes the observation results in different scenarios.

Table 3-29 SSL renegotiation results when the base station works as the SSL server

Value of SSL.RENEGO (LTE eNodeB, 5G gNodeB)	Value of SSL.RENEGO/INTERVAL (LTE eNodeB, 5G gNodeB)	Observation Result
DISABLE	Not configured	The results on the SSL server side are that the response types of SSL renegotiation are all DENY .
ENABLE	X (minutes). The value of X ranges from 1 to 60.	If the interval between two SSL renegotiation requests received by the SSL server is greater than X minutes, the response types of SSL renegotiation observed from the SSL server side may be SUCCESS or FAILURE but cannot be DENY . If the interval between the second SSL renegotiation request and the first SSL negotiation request is less than or equal to X minutes, the response types of SSL renegotiation observed from the SSL server side are all DENY .
ENABLE	0	The results on the SSL server side are that the response types of SSL renegotiation may be SUCCESS or FAILURE but cannot be DENY .

- SSL cipher suite configuration

A security scanning tool scans the ports transmitting messages between the SSL server and the SSL client. After an SSL connection is established, check the cipher suites carried in the ClientHello and ServerHello messages.

3.4.1.4 Configuring the OM Channel on the MAE

Use the SSL connection management function of the MAE to change the connection type and authentication method used between the NE and the MAE. The detailed procedure is as follows:

- Step 1** Log in to MAE and choose **Security > Certificate Authentication Management > SSL Connection Management** from the MAE-Access.
- Step 2** In the left pane, select the NE to be configured. In the right pane, set the connection type and authentication method, as shown in [Figure 3-14](#).

Figure 3-14 Changing the SSL configuration



NE Name ^	Connection Type ^	SSL Authentication ^	Connection Status ^
security-65	SSL	OSS authenticates NE	Connected

- Step 3** To check the status of an SSL connection between the base station and the MAE, select the base station in the SSL connection management window and then check the value of the **Connection Status** field. If the value of this field is **Connected**, an SSL connection has been successfully established.

----End

3.4.1.5 Network Monitoring

None

3.4.2 For the BSC6900/BSC6910 (GSM/UMTS)

3.4.2.1 When to Use

When the BSC6900/BSC6910 and the MAE are located on different networks, it is recommended that the SSL feature be activated to secure the OM channel between the BSC6900/BSC6910 and the MAE.

3.4.2.2 Data Configuration

3.4.2.2.1 Data Preparation

SSL Connection for the OM Channel

(Optional) Prepare data about the connection type supported by the BSC6900/BSC6910, as described in [Table 3-30](#).

Table 3-30 Connection type supported by the BSC6900/BSC6910

Parameter Name	Parameter ID	Setting Notes
Connection Type	CONNTYPE.CONNTYPE	The default value of this parameter is ONLY_SSL, which indicates that all application data transmitted over the TCP layer is protected by SSL. In this case, if the peer end does not support SSL, the communicating parties cannot establish a connection. If this parameter is set to ALL, all connection types, including SSL connection, are supported. In this case, the OM channel of the BSC6900/BSC6910 may not be protected by SSL, which imposes security risks. The default value is recommended.

Prepare data about the SSL authentication mode of the OM channel, as described in [Table 3-31](#).

Table 3-31 SSL authentication mode

Parameter Name	Parameter ID	Setting Notes
Authentication Mode	SSLAUTHMODE.AUTHMODE	Set this parameter based on the network plan. • If the SSL authentication is bidirectional, set this parameter to PEER. • If the SSL authentication is anonymous or if only the MAE authenticates the NE, set this parameter to NONE. The recommended value of this parameter is PEER.

(Optional) Prepare data about the SSL connection version of the OM channel, as described in [Table 3-32](#).

Table 3-32 SSL connection version

Parameter Name	Parameter ID	Setting Notes
TLS/SSL Version	SSLCONF.VERSION	If the operator has high security requirements, it is recommended that this parameter be set to TLSV12 .

Prepare the data about the certificates to authenticate the SSL connection of the OM channel. For details, see the "Data Preparation" section for PKI deployment on the base station controller in *PKI*.

BSC6900/BSC6910 Functioning as the FTP Client

Prepare data about the FTP client connection, as described in [Table 3-33](#).

Table 3-33 FTP client connection settings

Parameter Name	Parameter ID	Setting Notes
The Encrypted Mode	FTPSCLT.ENCYRMODE	The recommended value is ENCRYPTED .
Support State Firewall	FTPSCLT.SPTSTATEFW	Set this parameter based on the network plan.
Support SSL Certificate Authentication	FTPSCLT.SSLCERTAUTH	If this parameter is set to YES , the root certificate used on the FTP server must be preconfigured on the BSC6900/BSC6910. This root certificate is used by the BSC6900/BSC6910 to authenticate the certificate of the FTP server.

BSC6900/BSC6910 Functioning as the FTP Server

Prepare data about the FTP server connection, as described in [Table 3-34](#).

Table 3-34 FTP server connection settings

Parameter Name	Parameter ID	Setting Notes
Transport Encrypted Mode	FTPSSRV.ENCYRMODE	The recommended value is ENCRYPTED .

Parameter Name	Parameter ID	Setting Notes
The Type of FTP Server Command Port	FTPSSRV.DFTPORTS WT	Set this parameter to DEFAULTPORT or CUSTOMPORT .
The Command Port of FTP Server	FTPSSRV.SRVCMDPO RT	Set this parameter only when the FTPSSRV.DFTPORTSWT parameter is set to CUSTOMPORT .
The Source Data Port of FTP Server	FTPSSRV.SRVDATAP ORT	Set this parameter only when the FTPSSRV.DFTPORTSWT parameter is set to CUSTOMPORT .
Passive mode data port lower limit	FTPSSRV.ACDPORTL WLT	Set this parameter based on the network plan.
Passive mode data port upper limit	FTPSSRV.ACDPORTU PLT	Set this parameter based on the network plan.
TLS/SSL Version	FTPSSRV.SSLVER	Set this parameter to TLSV12 . If the base station controller is connected to the MAE or a third-party FTP client, ensure that the MAE or third-party FTP client is enabled with TLS1.2.

Policy for a Web LMT Login

Prepare data about the policy for a Web LMT login, as described in [Table 3-35](#).

Table 3-35 Setting the login policy of the Web LMT

Parameter Name	Parameter ID	Setting Notes
Policy for login to LMT and transmission	WEBLOGINPOLICY.P OLICY	The recommended value of this parameter is HTTPS .
TLS/SSL Version	WEBLOGINPOLICY.SS LVER	Set this parameter to TLSV12 . If a third-party browser is used to open the Web LMT, ensure that the browser is enabled with TLS1.2.

SSL Renegotiation Management

Prepare data about SSL renegotiation management (the corresponding MO is **SSLCONF**), as described in [Table 3-36](#).

Table 3-36 SSL renegotiation management

Parameter Name	Parameter ID	Setting Notes
Renegotiation Switch	SSLCONF.RENEGO	The recommended value of this parameter is ENABLE .
Renegotiation Interval	SSLCONF.RENEGOINTERVAL	The recommended value of this parameter is 60 .

SSL Cipher Suite Configuration

Prepare data about SSL cipher suites, as described in [Table 3-37](#).

Table 3-37 SSL cipher suite configuration

Parameter Name	Parameter ID	Setting Notes
Lowest Cipher Suite Level	SSLCONF.LOWESTCSLEVEL	It is recommended that this parameter be set to the same value for all NEs in an operator's network. - For security reasons, it is recommended that this parameter be set to HIGH for newly deployed SRAN10.1 NEs. - For an NE of a version earlier than SRAN10.1, this parameter is automatically set to LOW for compatibility considerations after the NE is upgraded to SRAN10.1.
Cipher Suite Name	SSLCS.CSNAME	Cipher suite names must comply with RFC 5246. Cipher suite names cannot be customized and must be defined before product delivery, as listed in Table 3-1 .

Parameter Name	Parameter ID	Setting Notes
Cipher Suite Level	SSLCS.CSLEVEL	It is recommended that the default security levels of cipher suites before delivery be used, as listed in Table 3-1. This parameter needs to be configured based on the actual situations. For example: • Set this parameter to DISABLE to disable a cipher suite. • If a cipher suite becomes a weak algorithm due to its security vulnerabilities, change its security level to LOW. NOTE For details about this parameter, see Table 3-2. If the operator has high security requirements, it is recommended that the following cipher suites whose Cipher Suite Level is set to MEDIUM be disabled. After these cipher suites are disabled, SSL3.0 cannot be used. In addition, certain functions of the Web LMT, such as tracing and monitoring, cannot be used. • TLS_RSA_WITH_3DES_EDE_CBC_SHA • TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA • TLS_DHE_DSS_WITH_3DES_EDE_CBC_SHA

3.4.2.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Configuring SSL for the OM channel  
//Setting the SSL authentication mode on the MAE
```

```
SET SSLAUTHMODE: AUTHMODE=PEER;
//Configuring the certificates used for SSL authentication
MOD APPCERT: APPTYPE=SSL,APPCERT="_ClientCer.pem";
//Setting the SSL version
SET SSLCONF: VERSION=TLSV11-1&TLSV12-1;
//Setting the security policy for the FTP client
SET FTPSCLT: ENCRYPMODE=Encrypted, SPTSTATEFWL=Yes, SSLCERTAUTH=Yes;
//Setting the security policy for the FTP server
SET FTPSSRV: ENCRYPMODE=ENCRYPTED,
DFTPORTSWT=DEFAULTPORT,ACDPORTLWLT=25000,ACDPORTUPLT=30000, SSLVER=TLSV11-1&TLSV12-1;
//Setting the login policy of the Web LMT
SET WEBLOGINPOLICY: POLICY=HTTPS, SSLVER=TLSV11-1&TLSV12-1;
//Configuring SSL renegotiation management
SET SSLCONF:RENEGO=ENABLE,RENEGOINTERVAL=60;
//Configuring the selection policy of cipher suites
SET SSLCONF:LOWESTCSLEVEL=HIGH;
//Setting the security levels of cipher suites (The security levels of cipher suites are set based on site
requirements. The following two scenarios are used as examples.)
```

- **Scenario 1**

//When the cipher suite ECDHE_ECDSA_AES_256_GCM_SHA384 must be disabled for an operator
SET SSLCS:CSNAME=ECDHE_ECDSA_AES_256_GCM_SHA384, CSLEVEL=DISABLE;

- **Scenario 2**

//When the cipher suite ECDHE_ECDSA_AES_256_GCM_SHA384 becomes a weak algorithm due to increasing computing capabilities
SET SSLCS:CSNAME=ECDHE_ECDSA_AES_256_GCM_SHA384, CSLEVEL=LOW;

3.4.2.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.4.2.3 Activation Verification

- **SSL for the OM channel**

In the SSL connection management window of the MAE, check whether the connection between the NE and the MAE is normal. If the connection is normal, SSL has been successfully activated on the OM channel.

- **FTP client**

On the MAE, set the transmission mode to FTPS and select the target BSC6900/BSC6910 in the navigation tree. On the license interface of the BSC6900/BSC6910, deliver a license file to the BSC6900/BSC6910. Check whether the license file is successfully transmitted. If yes, the FTPS function is successfully enabled.

- **FTP server**

Use the Web LMT offline tool to set the FTP client to FTPS mode. Log in to the FTP server of the BSC6900/BSC6910 and upload a test file (for example, test.txt) to the BSC6900/BSC6910. Check whether the test file is successfully uploaded. If yes, the FTPS function is successfully enabled.

- **Connection between the BSC6900/BSC6910 and the Web LMT**

Set the login policy of the Web LMT for the BSC6900/BSC6910 to HTTPS and log in to the BSC6900/BSC6910 from the Web LMT. A successful login to the BSC6900/BSC6910 indicates that an HTTPS connection has been successfully established between the BSC6900/BSC6910 and the Web LMT.

- SSL renegotiation management
The observation method is the same as that on the base station side. For details, see [3.4.1.3 Activation Verification](#).
- SSL cipher suite configuration
The observation method is the same as that on the base station side. For details, see [3.4.1.3 Activation Verification](#).

3.4.2.4 Configuring the OM Channel of the BSC6900/BSC6910 on the MAE

On the MAE, change the connection type and authentication method used between the BSC6900/BSC6910 and the MAE using the SSL connection management function of the MAE. The procedure is as follows:

- Step 1** Log in to the MAE and choose **Security > Certificate Authentication Management > SSL Connection Management** from the MAE-Access.
- Step 2** In the left pane, select the NE to be configured. In the right pane, set the connection type and authentication method, as shown in [Figure 3-15](#).

Figure 3-15 Changing the SSL configuration

SSL Connection Policy Management			
NE Name ^	Connection Type ^	SSL Authentication ^	Connection Status v
MBSC13	SSL Connection	NE Authentication OSS	Disconnected

----End

To check the status of an SSL connection on the MAE, do as follows:

Select the corresponding NE in the SSL connection management window and then check the value of the **Connection Status** field. If the value of this field is **Connected**, an SSL connection has been successfully established.

3.4.2.5 Network Monitoring

None

3.4.3 For the BSC6960 (GSM/UMTS)

3.4.3.1 When to Use

When the BSC6960 and the MAE are located on different networks, it is recommended that the SSL feature be activated to secure the OM channel between the BSC6960 and the MAE.

3.4.3.2 Data Configuration

3.4.3.2.1 Data Preparation

SSL Connection for the OM Channel

(Optional) Collect the connection type supported by the BSC6960 (the corresponding MML command is **SET CONNTYPE**), as described in [Table 3-38](#).

Table 3-38 Connection type

Parameter Name	Parameter ID	Setting Notes
Connection Type	SSL.CONNTYPE	The default value is SSL . It is recommended that this parameter be set to SSL to enable SSL-based security connection. If this parameter is set to ALL , all connection types are supported. In this case, the BSC6960 can be connected in non-SSL mode, and information transmitted in plaintext may be disclosed or tampered with.

Prepare data about the SSL authentication mode of the OM channel (the corresponding MML command is **SET SSLAUTHMODE**), as described in [Table 3-39](#).

Table 3-39 SSL authentication mode

Parameter Name	Parameter ID	Setting Notes
Authentication Mode	SSL.AUTHMODE	The default value of this parameter is PEER. Ensure that certificates are correctly configured on the MAE and BSC6960. It is recommended that this parameter be set to PEER to verify the MAE identity. If this parameter is set to NONE, the MAE identity cannot be determined. In this case, the MAE may be spoofed.

(Optional) Prepare data about the SSL connection version of the OM channel (the corresponding MML command is **SET SSLCONF**), as described in [Table 3-40](#).

Table 3-40 SSL connection version

Parameter Name	Parameter ID	Setting Notes
TLS/SSL Version	SSL.VERSION	TLS1.2 and TLS1.3 are enabled by default.

(Optional) Prepare certificate data, as described in [Table 3-41](#). The parameters in the following MO specify the device certificate used for SSL authentication.

Table 3-41 Certificate configuration

Parameter Name	Parameter ID	Setting Notes
Application Type	APPCERT.APPTYPE	Set this parameter to SSL .
Certificate File Name	APPCERT.APPCERT	Set this parameter based on the network plan. If an operator-issued device certificate is used for authentication, set this parameter to the name of the certificate.

(Optional) Prepare information about the trust group, as described in [Table 3-42](#). The trust group is used for peer verification in SSL authentication.

Table 3-42 Information about the trust group

Parameter Name	Parameter ID	Setting Notes
Trust Group ID	SSL.TRUSTGROUPID	This parameter is used for verifying the device certificates of the peer end. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the TRUSTGROUP MO are used to authenticate the peer end. If this parameter is set to 65535 , no trust group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.

(Optional) Configure the TLS configuration policy for the OM channel, as described in [Table 3-43](#).

Table 3-43 TLS configuration policy for the OM channel

Parameter Name	Parameter ID	Setting Notes
Policy ID	SSL.POLICYID	The default value of this parameter is 0 . For the TLS configuration policy numbered 0, the key update function is enabled, the key update period is 60 minutes, and all TLS1.3 cryptographic algorithms are enabled. Before setting this parameter, ensure that the corresponding TLS configuration policy has been configured. If the corresponding TLS configuration policy does not exist, configure a TLSPOLICY MO and then set this parameter to the corresponding policy ID.

FTP Client Connection

Prepare data about the FTP client connection, as described in [Table 3-44](#).

Table 3-44 FTP client connection data

Parameter Name	Parameter ID	Setting Notes
Transport Encrypted Mode	FTPCLT.ENCYRMODE	It is recommended that this parameter be set to Encrypted to enable FTPS. If this parameter is set to Plaintext , data will be transmitted in plaintext over FTP connections, which may cause information leakage and tampering.
Support State Firewall	FTPCLT.SPTSTATEFW L	Set this parameter based on whether the stateful firewall is deployed on the customer's live network.

Parameter Name	Parameter ID	Setting Notes
Support SSL Certificate Authentication	FTPCLT.SSLCERTAUTH	It is recommended that this parameter be set to Yes to authenticate the FTP server certificate. If this parameter is set to No, the peer certificate is not authenticated and whether the FTP server identity is valid cannot be determined. As a result, the FTP server may be spoofed. In this version, the setting of this parameter can still be synchronized and delivered through the interface, and the setting can still take effect. In later versions, however, this parameter will be disused. Therefore, avoid using this parameter. To prevent incorrect configurations, you can run the SET SECPLY command to adjust the configuration security level. The No value is not allowed to be used.
Policy ID	FTPCLT.POLICYID	The default value of this parameter is 0. For the TLS configuration policy numbered 0, the key update function is enabled, the key update period is 60 minutes, and all TLS1.3 cryptographic algorithms are enabled. Before setting this parameter, ensure that the corresponding TLS configuration policy has been configured. If the corresponding TLS configuration policy does not exist, configure a TLSPOLICY MO and then set this parameter to the corresponding policy ID.
Trust Group ID	FTPCLT.TRUSTGROUPID	This parameter is used for verifying the device certificates of the peer end. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the TRUSTGROUP MO are used to authenticate the peer end. If this parameter is set to 65535, no trust group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.

SSL Renegotiation Management

Prepare data about SSL renegotiation management (the corresponding MML command is **SET SSLCONF**), as described in [Table 3-45](#).

Table 3-45 SSL renegotiation management

Parameter Name	Parameter ID	Setting Notes
Renegotiation Switch	SSL.RENEGO	The recommended value of this parameter is ENABLE .
Renegotiation Interval	SSL.RENEGOINTERVAL	The recommended value of this parameter is 60 .

NOTE

The SSL renegotiation management configuration does not take effect for TLS1.3.

SSL Key Update Management

Prepare data about SSL key update management, as described in [Table 3-46](#).

Table 3-46 SSL key update management

Parameter Name	Parameter ID	Setting Notes
POLICY ID	TLSPOLICY.POLICYID	This parameter is used together with: • SSL.POLICYID to specify the TLS configuration policy used when SSL is applied to the OM channel. • CA.POLICYID to specify the TLS configuration policy used when SSL is applied to CMPv2-based interaction scenarios. • FTPCLT.POLICYID to specify the TLS configuration policy used when SSL is applied to FTP applications. • CRLTSK.POLICYID to specify the TLS configuration policy used when SSL is applied to LDAP applications.
Key Update Switch	TLSPOLICY.KEYUPDATESW	The default value ON is recommended. If this parameter is set to OFF , the key may not be updated for a long time and may be cracked.
Key Update Lifetime	TLSPOLICY.KEYUPDTEL	The default value is recommended.

SSL Cipher Suite Configuration

Prepare data about TLS1.2 cipher suites, as described in [Table 3-47](#).

Table 3-47 SSL cipher suite configuration

Parameter Name	Parameter ID	Setting Notes
Lowest Cipher Suite Level	<i>SSL.LOWESTCSLEVEL</i>	It is recommended that this parameter be set to the same value for all NEs in an operator's network. It is recommended that this parameter be set to HIGH to use an algorithm with a high security level. If this parameter is set to MEDIUM or LOW, some weak encryption algorithms are compatible, which may be cracked. It is recommended that this parameter be set to HIGH for newly deployed NEs.
Cipher Suite Name	<i>SSLCS.CSNAME</i>	This parameter specifies the name of a cipher suite, which complies with RFC 5246. Cipher suite names cannot be customized and must be defined before product delivery, as listed in Table 3-1.

Parameter Name	Parameter ID	Setting Notes
Cipher Suite Level	SSLCS.CSLEVEL	This parameter specifies the security level of a cipher suite. It is recommended that this parameter retain its default value, as listed in Table 3-1. This parameter needs to be configured based on the actual situations. For example: - Set this parameter to DISABLE to disable a cipher suite. - If a cipher suite becomes a weak algorithm due to its security vulnerabilities, change its security level to LOW. For details, see Table 3-2. If the value of this parameter is changed from HIGH to MEDIUM, LOW, or DISABLE, both sides may fail to negotiate a cipher suite and consequently connections cannot be established. Therefore, exercise caution when setting this parameter.

[Table 3-48](#) describes the parameters that must be set in the **TLSPOLICY** MO to configure TLS1.3 cipher suites.

Table 3-48 TLS1.3 cipher suite configuration

Parameter Name	Parameter ID	Setting Notes
Diffie-Hellman Group	TLSPOLICY.DHGRP	By default, all algorithms are enabled. The algorithm SECP256R1 cannot be disabled. Set this parameter based on the network plan.
Signature Algorithm	TLSPOLICY.SIGALG	By default, all algorithms are enabled. The algorithm RSA_PSS_RSAE_SHA256 cannot be disabled. Set this parameter based on the network plan.
Cipher Suite	TLSPOLICY.CIPHERSUITE	By default, all algorithms are enabled. The algorithm TLS_AES_128_GCM_SHA256 cannot be disabled. Set this parameter based on the network plan.

Lists of Cipher Suites Supported by the BSC6960 Functioning as the Client

- FTP over SSL

A BSC6960 supports TLS1.2 and TLS1.3 when it functions as the FTPS client. The following table lists the SSL cipher suites supported by the BSC6960.

No.	SSL Cipher Suite
1	ECDHE_ECDSA_AES_256_GCM_SHA384
2	ECDHE_RSA_AES_256_GCM_SHA384
3	ECDHE_ECDSA_AES_128_GCM_SHA256
4	ECDHE_RSA_AES_128_GCM_SHA256

- HTTP over SSL

A BSC6960 supports TLS1.2 and TLS1.3 when it functions as the CMPv2 client. The following table lists the SSL cipher suites supported by the BSC6960.

No.	SSL Cipher Suite
1	ECDHE_ECDSA_AES_256_GCM_SHA384
2	ECDHE_RSA_AES_256_GCM_SHA384
3	ECDHE_ECDSA_AES_128_GCM_SHA256
4	ECDHE_RSA_AES_128_GCM_SHA256

- LDAP over SSL

The following table lists the cipher suites supported when StartTLS for LDAP is applied and authenticating the certificate of the peer end is required.

No.	Cipher Suite with Minimum DH Length Being DH_0	Cipher Suite with Minimum DH Length Not Being DH_0
1	ECDHE_ECDSA_AES_256_GCM_SHA384	ECDHE_ECDSA_AES_256_GCM_SHA384
2	ECDHE_RSA_AES_256_GCM_SHA384	ECDHE_RSA_AES_256_GCM_SHA384
3	ECDHE_ECDSA_AES_128_GCM_SHA256	ECDHE_ECDSA_AES_128_GCM_SHA256
4	ECDHE_RSA_AES_128_GCM_SHA256	ECDHE_RSA_AES_128_GCM_SHA256

3.4.3.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Configuring SSL for the OM channel
//Configuring the type of device certificate used for SSL authentication
MOD APPCERT: APPTYPE=SSL, APPCERT="appcert.pem";
//Setting the connection policy for the FTP client
SET FTPSCLT: ENCRYPMODE=Encrypted, SPTSTATEFWL=Yes, SSLCERTAUTH=Yes, TRUSTGROUPID=65535;
//Setting the SSL version for the OM channel
SET SSLCONF: VERSION=TLSV12-1&TLSV13-1;
//Setting the TLS configuration policy for the OM channel
SET SSLCONF: POLICYID=0;
//Setting a trust group for the OM channel
SET SSLCONF: TRUSTGROUPID=65535;
//Configuring SSL renegotiation management
SET SSLCONF: RENEGO=ENABLE, RENEGOINTERVAL=60;
//Configuring key update management
MOD TLSPOLICY: POLICYID=0, KEYUPDATESW=ON, KEYUPDATELT=60;
//Configuring the selection policy of cipher suites
SET SSLCONF:LOWESTCSLEVEL=HIGH;
//Setting the security levels of cipher suites (The security levels of cipher suites are set based on site requirements. The following two scenarios are used as examples.)
```

- **Scenario 1:**
//When the cipher suite ECDHE_ECDSA_AES_256_GCM_SHA384 must be disabled for an operator
SET SSLCS:CSNAME=ECDHE_ECDSA_AES_256_GCM_SHA384, CSLEVEL=DISABLE;
- **Scenario 2:**
//When the cipher suite ECDHE_ECDSA_AES_256_GCM_SHA384 becomes a weak algorithm due to increasing computing capabilities
SET SSLCS:CSNAME=ECDHE_ECDSA_AES_256_GCM_SHA384, CSLEVEL=LOW;

3.4.3.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.4.3.3 Activation Verification

- **SSL for the OM channel**
In the SSL connection management window of the MAE, check whether the connection between the NE and the MAE is normal. If the connection is normal, SSL has been successfully activated on the OM channel.
- **FTPS**
Check whether log files are being transmitted between the BSC6960 and the MAE based on FTPS as expected. If log file transmission is normal, an FTPS connection has been successfully established between the BSC6960 and the MAE.
- **Connection between the BSC6960 and the Web LMT**
TLS1.2 and TLS1.3 are configured for the Web LMT by default. Log in to the BSC6960 through the Web LMT. A successful login to the BSC6960 indicates

that a connection has been successfully configured between the BSC6960 and the Web LMT.

- SSL renegotiation management

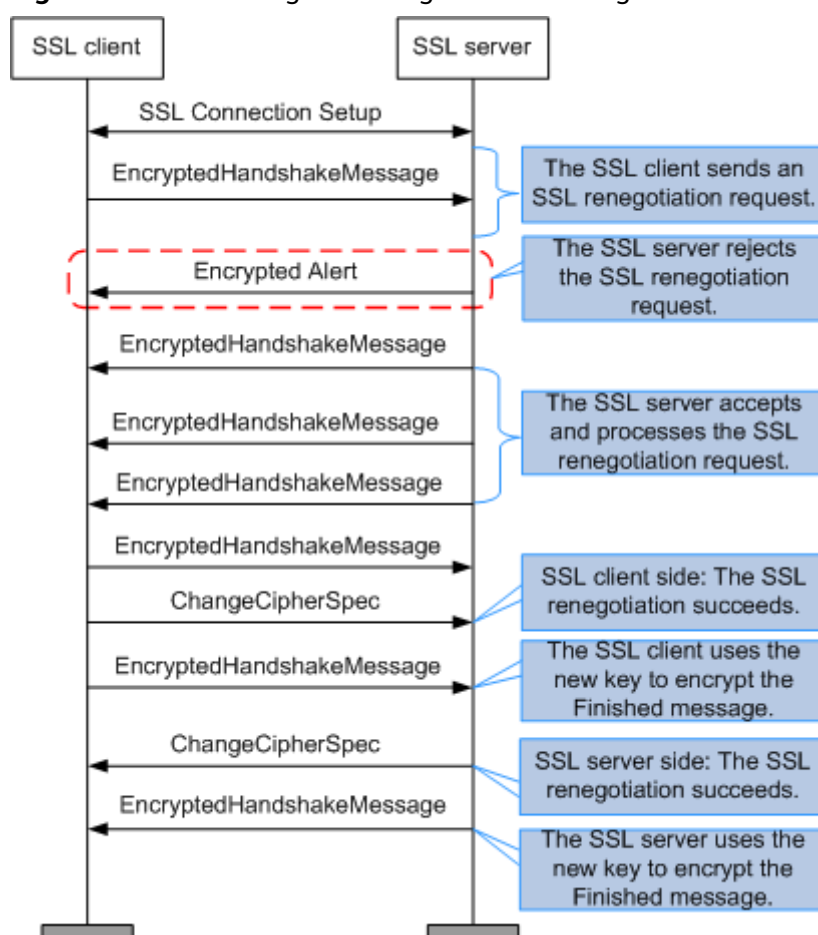
Method 1:

SSL renegotiation takes place between the SSL server and the SSL client. Therefore, a method for checking whether the SSL renegotiation management function works properly is to capture packets transmitted between the SSL server and the SSL client. The following uses the BSC6960 working as the SSL server as an example to describe how to observe SSL renegotiation management.

Prerequisites:

An SSL connection has been established between the SSL server and the SSL client. Then, enable the SSL client to send an SSL renegotiation request message to the SSL server.

Figure 3-16 Observing SSL renegotiation management



Analyze the messages transmitted between the SSL server and SSL client shown in [Figure 3-16](#):

- If the SSL client sends an EncryptedHandshakeMessage to the SSL server immediately after an SSL connection is established, this message is an encrypted Client Hello handshake message, indicating that the SSL client initiates an SSL renegotiation procedure.

- Upon receipt of the EncryptedHandshakeMessage from the SSL client, if the SSL server consecutively sends two to four EncryptedHandshakeMessage messages to the SSL client, the SSL server has accepted and processed the SSL renegotiation request. Among these message interactions, if both the SSL server and the SSL client send a ChangeCipherSpec message to the peer end, the SSL renegotiation procedure is completed, and the new cipher keys are used for communication.
- Upon receipt of the EncryptedHandshakeMessage from the SSL client, if the SSL server responds with an Encrypted Alert message, the SSL server has rejected the SSL renegotiation request.

Method 2:

Run the MML command **DSP SSLRENEGO** to check the value of the **Response Type** parameter. Based on the value of this parameter, the performance of the SSL renegotiation management function in reducing resource consumption and preventing DoS attacks will be evaluated.

The values of **Response Type** include **SUCCESS**, **DENY**, and **FAILURE**.

- **SUCCESS** indicates that the SSL server has accepted and processed the SSL client's renegotiation request and the SSL renegotiation is successful.
- **DENY** indicates that the SSL renegotiation management function is enabled at the SSL server and the SSL server has rejected the SSL client's renegotiation request.
- **FAILURE** indicates that the SSL server fails to complete the SSL renegotiation procedure due to exceptions not caused by the SSL renegotiation management function.

Table 3-49 describes the observation results in different scenarios.

Table 3-49 SSL renegotiation results when the BSC6960 works as the SSL server

Value of <i>SSL.RENEGO</i>	Value of <i>SSL.RENEGO/INTERVAL</i>	Observation Result
DISABLE	Not configured	The results on the SSL server side are that the response types of SSL renegotiation are all DENY .

Value of <i>SSL.RENEGO</i>	Value of <i>SSL.RENEGO/INTERVAL</i>	Observation Result
ENABLE	X (minutes). The value of X ranges from 1 to 60.	If the interval between two SSL renegotiation requests received by the SSL server is greater than X minutes, the response types of SSL renegotiation observed from the SSL server side may be SUCCESS or FAILURE but cannot be DENY . If the interval between the second SSL renegotiation request and the first SSL negotiation request is less than or equal to X minutes, the response types of SSL renegotiation observed from the SSL server side are all DENY .
ENABLE	0	The results on the SSL server side are that the response types of SSL renegotiation may be SUCCESS or FAILURE but cannot be DENY .

- SSL cipher suite configuration

A security scanning tool scans the ports transmitting messages between the SSL server and the SSL client. After an SSL connection is established, check the cipher suites carried in the ClientHello and ServerHello messages.

3.4.3.4 Configuring the OM Channel of the BSC6960 on the MAE

Use the SSL connection management function of the MAE to change the connection type and authentication method used between the NE and the MAE. The detailed procedure is as follows:

- Step 1** Log in to the MAE and choose **Security > Certificate Authentication Management > SSL Connection Management** from the MAE-Access.
- Step 2** In the left pane, select the NE to be configured. In the right pane, set the connection type and authentication method, as shown in [Figure 3-17](#).

Figure 3-17 Changing the SSL configuration



SSL Connection Policy Management			
NE Name ^	Connection Type ^	SSL Authentication ^	Connection Status ^
security-65	SSL	OSS authenticates NE	Connected

- Step 3** To check the status of an SSL connection between the BSC6960 and the MAE, select the corresponding NE in the SSL connection management window and then check the value of the **Connection Status** field. If the value of this field is **Connected**, an SSL connection has been successfully established.

----End

3.4.3.5 Network Monitoring

None

4 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

5 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

7 Reference Documents

1. IETF RFC 6101
2. IETF RFC 2246
3. IETF RFC 4346
4. IETF RFC 5246
5. *PKI* for SingleRAN
6. *SRAN Networking and Evolution Overview* for SingleRAN
7. *OM Security* for SingleRAN